

Visual Effects Shot

BSc Games and Multimedia Environments

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Introduction

Visual effects allow filmmakers to extend the scope of their imagination beyond what is possible with practical special effects and on-set methods of capturing a scene. They have allowed writers to delimit themselves while conjuring fiction from their minds and have permitted artists to conceive fantastical worlds with no bounds (*Figure 1*). On the other end of the spectrum, visual effects have allowed us to see what long extinct creatures (*Figure 2*), or ruined castles, or ancient leaders may have looked like in their time.

Most digital productions will use visual effect to a greater or lesser degree, be it in colour grading in post-production, chromakey (green screen) technology or lavish effects-driven visuals



Figure 1 Avatar: Way of the Water (Jacobs, 2022)



Figure 2 Prehistoric Planet (Anon., n.d.)

The development, improvement, increased accessibility and lowering costs of visual effects has had an impressive impact on the filmmaking industry, allowing productions to tell stories that they would not have dreamed of decades ago, from small budget films to heavily funded productions. The hope for this project is to create a visual effect shot on a low budget and this report will document the planning and creative processes that are involved.

Aims

In this project, the aim is to make a short video sequence containing a variety of production techniques including shot composition, lighting, sound, set dressing, green screens and costumes and post-production techniques including colour correction, rotoscoping, 3D modelling, texturing, virtual lighting, and compositing.

The sequence will be made up of real and computer-generated elements and will be a representation of a seismic event which causes a crack in the earth to appear and grow. The shot should be as visually realistic as possible so that the viewer is fully immersed in the content.

The sequence, which will be about 20 seconds long, could be part of a disaster movie like 'San Andreas' or 'The Quake'.

Primary Objectives

- Discover what details make visual effects convincing to the audience by analysing successful and unsuccessful visual effects shot created by studios in the filmmaking industry.
- Draw a storyboard for the scene so that the footage can be shot in as few takes as possible, saving time.
- Create a pre-visualisation of the scene as a plan for shooting the footage and rendering the 3D elements.
- Take an HDRI on set so that the lighting can be matched in the software, leading to more realistic highlights and shadows.
- Model the required assets (buildings, debris, foliage etc.) so that they can be composited onto the footage.
- Create procedural textures to apply to the 3D assets.
- Camera track the footage.

Extensions

- Use the physics built into 3D software to simulate destruction on a larger scale.

- Create an explorable version of the shot using a games engine.
- Expand the shot with suitable audio and sound effects.

Project Relevance

This project will require extensive 3D modelling, animation, video editing and compositing ability which has been developed throughout the course in the '3D Modelling and Animation', 'Video Production Techniques', 'Programming for 3D' and 'Visual Effects' modules in years two and three. These skills will also require further improvement during the project. Some previously acquired experience in creating scenes in 3D space will be useful for developing a project in this area.

Professional Considerations

BCS Code of Conduct

"1. Public Interest

You shall:

- a. have due regard for public health, privacy, security and wellbeing of others and the environment.*
- b. have due regard for the legitimate rights of Third Parties.*
- c. conduct your professional activities without discrimination on the grounds of sex, sexual orientation, marital status, nationality, colour, race, ethnic origin, religion, age or disability, or of any other condition or requirement.*
- d. promote equal access to the benefits of IT and seek to promote the inclusion of all sectors in society wherever opportunities arise."*

To abide by this section, all risks of filming in studio and on location will be identified beforehand and a formal risk assessment will be written. Participants will be required to sign a consent form, acknowledging that they understand the perils.

The personal information of all persons included in the footage will not be disclosed and their faces will not be shown without their permission. Actors will not be asked to perform any tasks that they are not comfortable with or say anything they are not inclined to.

"2. Professional Competence and Integrity

You shall:

- a. only undertake to do work or provide a service that is within your professional competence.*
- b. NOT claim any level of competence that you do not possess.*
- c. develop your professional knowledge, skills and competence on a continuing basis, maintaining awareness of technological developments, procedures, and standards that are relevant to your field.*
- d. ensure that you have the knowledge and understanding of Legislation * and that you comply with such Legislation, in carrying out your professional responsibilities.*
- e. respect and value alternative viewpoints and, seek, accept and offer honest criticisms of work.*
- f. avoid injuring others, their property, reputation, or employment by false or malicious or negligent action or inaction.*
- g. reject and will not make any offer of bribery or unethical inducement."*

Having discussed with the technical supervisor, the decision has been made that it is suitable and completable.

This project will comply with the legislation.

Feedback will be obtained on the project weekly from the technical supervisor and peers who are doing related projects with the same supervisor. Criticisms of project related work will be accepted and offered, and the advice will be used to improve the project.

Care will be taken to ensure that the project is completed without affecting others by ensuring that all relevant parties are happy with their role in the project.

No bribery or unethical inducement will be accepted or offered.

“3. Duty to Relevant Authority

You shall:

- a. carry out your professional responsibilities with due care and diligence in accordance with the Relevant Authority’s requirements whilst exercising your professional judgement at all times.*
- b. seek to avoid any situation that may give rise to a conflict of interest between you and your Relevant Authority.*
- c. accept professional responsibility for your work and for the work of colleagues who are defined in a given context as working under your supervision.*
- d. NOT disclose or authorise to be disclosed, or use for personal gain, or to benefit a third party, confidential information except with the permission of your Relevant Authority, or as required by Legislation.*
- e. NOT misrepresent or withhold information on the performance of products, systems or services (unless lawfully bound by a duty of confidentiality not to disclose such information), or take advantage of the lack of relevant knowledge or inexperience of others.”*

The requirements of the Relevant Authority, the School of Informatics at the University of Sussex, will be met by being punctual while accomplishing work.

Conflicts of interest with the University will be avoided by discussing the project’s intentions with the technical supervisor.

Responsibility for the project and the work of those under the supervision of the project will be accepted.

No confidential information will be disclosed without the permission of the University.

This project will be honest about its progress and will not take advantage of others.

“4. Duty to the Profession

You shall:

- a. accept your personal duty to uphold the reputation of the profession and not take any action which could bring the profession into disrepute.*
- b. seek to improve professional standards through participation in their development, use and enforcement.*
- c. uphold the reputation and good standing of BCS, the Chartered Institute for IT.*
- d. act with integrity and respect in your professional relationships with all members of BCS and with members of other professions with whom you work in a professional capacity.*
- e. encourage and support fellow members in their professional development.” (BCS, n.d.)*

This project will be completed in accordance with the BCS Code of Conduct, and it will be completed professionally.

All members of the BCS will be respected and uplifted.

Ethics

As the project will involve filming on location. The risks associated with this will be managed. Precautions will be taken so that the project complies with the User Testing Compliance Form attached.

Requirements Analysis

The project should portray the shift of tectonic plates on a large scale, leaving a deep ravine through the landscape. The project should contain one shot of high detail and is sufficiently realistic as to be a convincing VFX rendition of reality.

The project should utilise advanced software tools, including any required add-ons, and high-end computer hardware to successfully execute the VFX shot with the desired effect.

The project will require a certain level of skill and expertise necessary to develop the desired shot.

The project must be planned carefully, and time must be managed accordingly to achieve the goals of the project.

The target group of users for this project are the viewers who will see the shot. Ultimately, their needs are to be entertained and so the aim is to create an engaging and thrilling sequence.

Similar Shots

San Andreas (2015)

This film depicts a series of earthquakes and subsequent disasters that occur along the San Andreas fault in California. One shot shows the aftermath of an earthquake where the ground has split in two. The ending of the shot would be similar to this. The shot will show the process of how the landscape comes to be in this state, with the crack starting at a few millimetres and expanding all the way to several meters wide.



Figure 3 *San Andreas (2015)* directed by Brad Peyton earthquake aftermath, VFX by Image Engine.

2012 (2009)

The supermarket scene from the disaster movie 2012 features many of the elements that will be used in the project. In the beginning of this scene, small cracks in the floor appear that end up developing to what we see in Figure 4. It includes many physics simulations of the supermarket items and parts of the floor falling into the ground, something that will be used in the project.



Figure SEQ Figure 1* ARABIC 4 2012 (2009) directed by Roland Emmerich supermarket scene, VFX by Uncharted Territory.

Development Process

Before commencing with the creative process of making the shot, research has been conducted into what will make it successful. Existing examples, along with behind-the-scenes explanations of how they were created have been watched so that the techniques used can be understood in more detail.

This section will discuss the technical specifications that must be considered before commencing with the production of a visual effects shot.

Although the aim is to create a visually realistic shot, the project does not intend to simulate in a scientifically accurate way a real seismic event. For this reason, extensive research into the effects of earthquakes is not necessary. Cracks formed by earthquakes are usually much shallower than those seen in the disaster films in the previous section and do not happen at all on diverging boundaries (where tectonic plates move away from each other) (Adams, 2015). The shot will be a dramatized, exaggerated version of the destruction an earthquake can cause. However, recent events in Turkey and Syria have shown that similar situations can be seen in nature. This [video](#) (ITV News, 2023) shows a real-world example.

Simulating destruction in 3D can be difficult as it requires a lot of processing power and time. Blender will be used for the majority of the 3D workflow as it has been used before for personal use and provides the flattest learning curve compared to Cinema 4D for example. As it is an open source, free software, there are many add-ons and tutorials on how to use their features. This will be helpful during the development of the shot as it will allow the project to be pushed further in terms of complexity.

The 'Cell fracture' add-on makes it easier to destroy objects in Blender. It can be used to split objects into smaller pieces so that they can fall into the crack when physics are applied.

Video

Resolution

The resolution of a video is an important factor which dictates how clear the image will look. There are several standard resolutions used which are measured by the number horizontal and vertical pixels in the frame. The total number of pixels in the image can then be calculated by multiplying these numbers. The higher the number of pixels, the higher the resolution and the higher the clarity

of the image. Higher resolutions will also require more storage space as the files contain more information (Leonard & Kurniawan, n.d.).

Although still considered High Definition (HD), the 720p (1280x720 pixels) resolution has largely been made obsolete by 1080p. This is a result of higher resolution displays becoming standard across all forms of media.

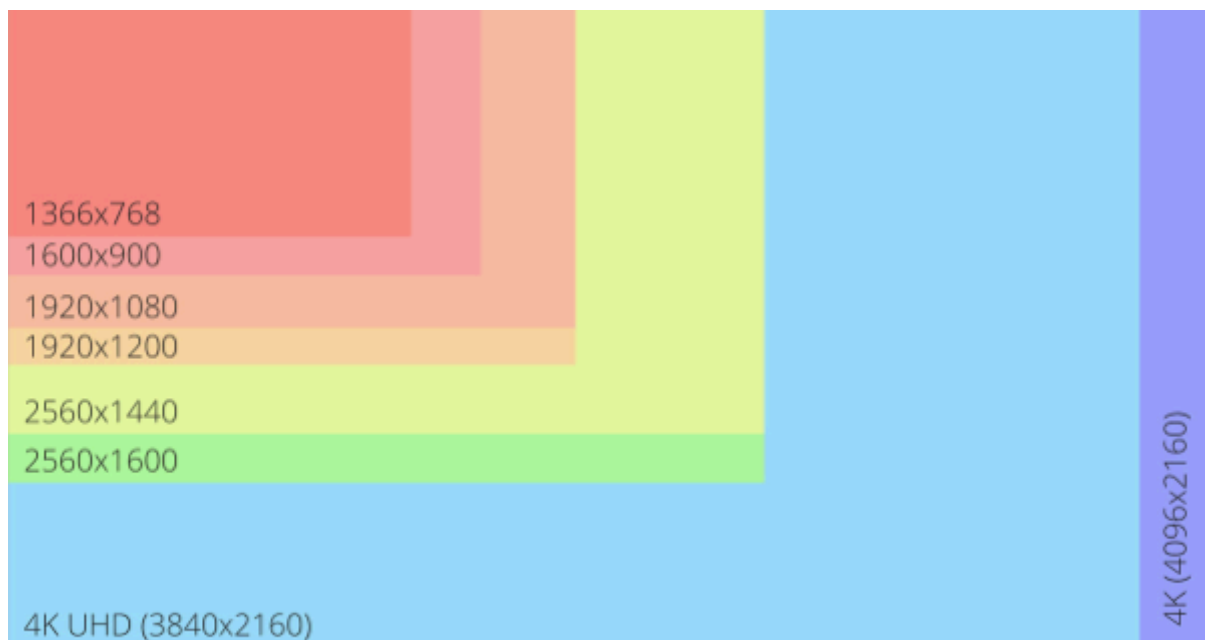


Figure SEQ Figure * ARABIC 5 Chart comparing resolutions.

1080p (1920x1080 pixels), also known as full HD, is the most used resolution as it offers a detailed image, while keeping the storage requirements low.

Slightly above 1080p is the 2K (2048x1080 pixels) resolution which can be seen used in most cinema screens. This resolution uses a different Aspect Ratio to 1080p as it is wider (Darlington, n.d.).

Another 2K resolution is 1440p (2560x1440 pixels). Also known as Quad HD or QHD, it provides a resolution that serves as a middle ground between 1080p and 4K.

4K UHD (3840x2160 pixels) allows artists in post-production to crop and zoom on the footage while maintaining relatively high quality. If, for example, the editor was exporting at 1080p but using 4K footage, they would be able to use as little as 25% of the screen without losing fidelity. This grants artists the ability to manipulate the image in more ways. In order to maintain the highest level of detail and being able to zoom in on the footage if required, this resolution was used for the project

Overall, choosing a resolution comes down to what the video will be used for and what displays it will be shown on. High resolution images on a low-resolution display will not be fully utilised but low-resolution images on a high-resolution display will look blurry and pixelated.

Frame rate

Video is recorded by taking multiple images in a short space of time. Frame rate describes the number of individual images, called frames, contained in one second of video. When humans watch video, it appears as if objects move smoothly when really the human eye is making up for the difference in each frame. As frame rates increase, the smoothness of the footage increases (Kurniawan & Hara, n.d.).

24fps is the typical frame rate for movies and television used in North and South America under the NTSC video standard. 25 fps is used in the UK and Europe and most other regions around the world due to the PAL standard (Heinzman, 2019). It provides a cinematic look and it's favoured by filmmakers. It was originally used to keep the cost of film down while being fast enough to create fluid motion for the human eye but is now used even in digital video.

30fps is commonly used in the film industry for content that is intended for television viewing. This could be in sports or news because there are more frames to capture movement. It was previously used in video games as it was an achievable frame rate on cheaper hardware but as gaming consoles and PCs improved, it is now considered slow for gaming. This frame rate was used for the project because it is the frame rate that the camera shot at and is suitable for displaying on computer screens and mobile devices.

60fps is also used in sports broadcasting and for video games because it can capture fast moving action and fluid motion. It reduces motion blur for the viewer and leads to a more comfortable viewing experience. Most video gaming platforms now can output 60fps at a resolution of 1080p or higher. The movie industry tends not to use frame rates this high because viewers find that it looks too realistic and takes away from the narrative of the cinematic experience and looks too much like a video game like where watchers of The Hobbit Trilogy found the experience uncomfortable (Kastrenakes, 2014).

There are instances where higher frame rates can work in movies. Avatar 2 (2022) was filmed at 48fps for some of its scenes where a lot of movement happens. Director James Cameron told Yahoo News, "The rule was that whenever they're underwater, it's 48 frames. Boom. Don't even think about it. Some of the flying scenes and some of the broad vistas benefit from 48 frames." (Beasley, 2022). The higher frame rates work for these sequences partially because there is so much CGI that it already looks like a video game, though some viewers may still not enjoy it.

Frame rates higher than 60fps are usually reserved for video gaming but also are used when filmmakers want to shoot a slow-motion sequence as the footage can be slowed down while still maintaining a watchable frame rate.

The refresh rate of a display also impacts how a video is viewed. Refresh rate is the frequency that a display or projector updates the image on screen. If this is lower than the frame rate of the content, the display will not be able to show all the frames and information will be lost.

Aspect Ratio

The proportional relationship between a screen's width and height is referred to as the aspect ratio. A 16:9 aspect ratio, for instance, indicates that the image is 16 pixels wide for every 9 pixels tall. Filmmakers use different aspect ratios for different applications (Adobe, n.d.). This aspect ratio was used for the project as it is the most widely available for consumers.

The 4:3 aspect ratio was the most common aspect ratio in the past before widescreen televisions were released. It is now used stylistically when footage is meant to appear old.

16:9 is now used for all sorts of content. Most computer screens and TVs use it. A portrait version of it is used for many mobile phones to in the form of 9:16.

1.85:1 is the standard for cinema as it is slightly wider than 16:9. It can still be viewed on 16:9 screens but will be shown with black bars at the top or bottom of the screen.

The main factors that determine what aspect ratio to use is the platform or platforms it will be distributed on, the artistic goals of the project, and the audience.

[This resource](#) by Firehouse lists many resolutions and their aspect ratios.

The first step in the process was filming the environment that the earthquake would take place in. A wide, open, flat field was chosen for this as it would be easier to composite the simulated elements into a flatter space rather than one with variety in its height. It was important to this particular project to find an environment with few features like fences or livestock as they would have to be modelled in 3D and added to the physics simulations or removed from the footage entirely and that would require extra steps like clone tracking (Robinson, 2022) and that would take more time.

3D Tracking

To integrate computer-generated imagery with live action footage, motion tracking must be used. The procedure entails monitoring the movement of objects or actors inside a video to apply visual effects, such as compositing, 3D models, or special effects,

The first step in the motion tracking procedure is to analyse a video clip frame-by-frame to locate reference points, such as the corners or edges of the tracked objects. Points of high contrast and consistent shape work best. The programme then uses algorithms to determine these reference points' location, rotation, and scale as they move across the video.

Once the tracking information has been gathered, it can be utilised to build a virtual camera that moves similarly to the real camera that was used for filming. This enables VFX artists to produce and work with CGI components that mimic the movement and viewpoint of the live-action material.

Motion graphics and visual effects can be added to select segments of the film using motion tracking. A motion-tracked mask, for instance, can be used to isolate a figure or object in the footage so that visual effects can be used only there.

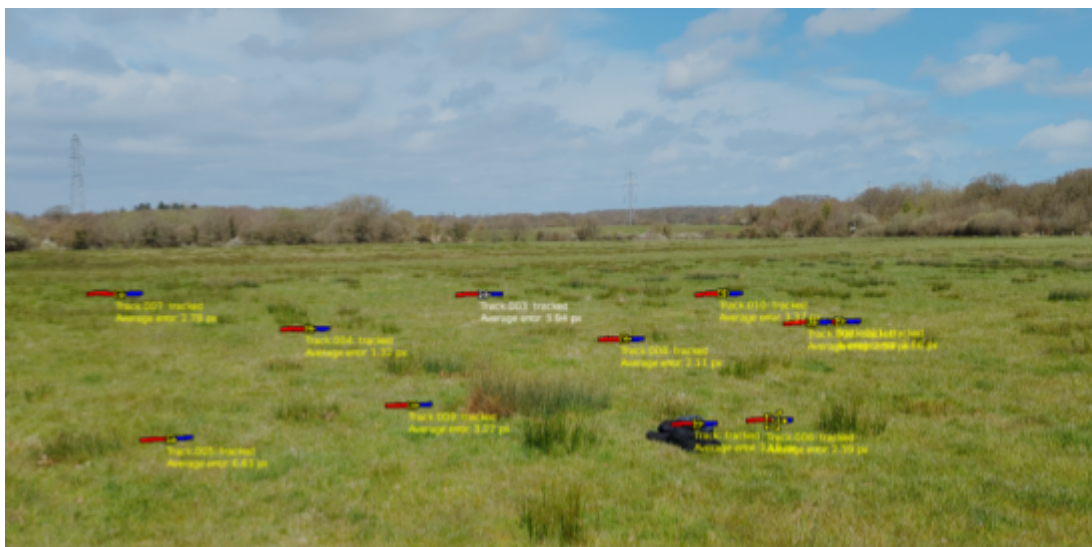


Figure SEQ Figure 1* ARABIC 6 Tracking points on footage.

Tracking of the footage in this project was done in Blender as importing usable track data from a compositing software like After Effects requires additional steps and is not always possible.

Tracking markers were placed on the footage at points of high contrast, and where the pixels were visible throughout the video. The objects placed in frame beforehand were useful as the markers attached to them were set as the ground plane for the footage. A low margin of error is required for the tracking data to be successful as it increases the similarity between the movements of the virtual camera and the footage. This meant that applying the track and putting the tracked camera into the scene was seamless and quite accurate without further adjustment, although some rotation of the

camera was required so that the top layer would line up with the image but after that the track was complete.

3D Models

Polygonal Modelling

3D modelling works by software creating points in space that have coordinates along the x, y, and z axes. These points, known as vertices, can be connected with edges to create lines in 3D space. If 3 or more vertices are connected to each other, a face is created. Modelling using a combination of vertices, edges, and faces is known as polygonal modelling. Objects are expanded by adding additional geometry and connecting them together.

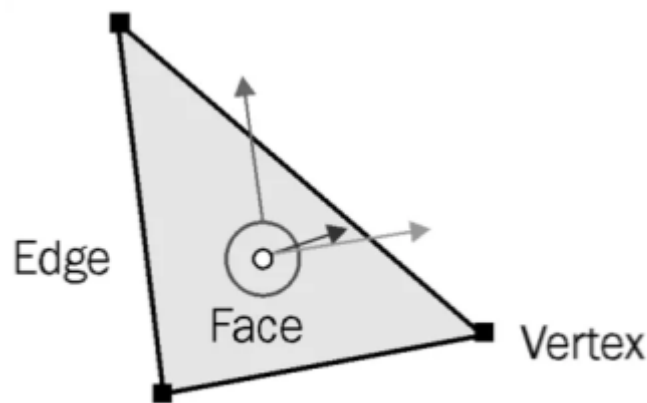


Figure SEQ Figure * ARABIC 7 Diagram labelling vertices, edges, and facesCITATION pac12 V 2057 (packt, 2012).

NURBS Modelling

Non-uniform rational basis splines (NURBS) modelling uses curves known as basis splines to create shapes instead of polygons. NURBS use control points along the spline that are weighted to determine the shape of the curve. It is used in engineering as it allows for a degree of accuracy that would require many polygons to recreate (Tan, 2021).

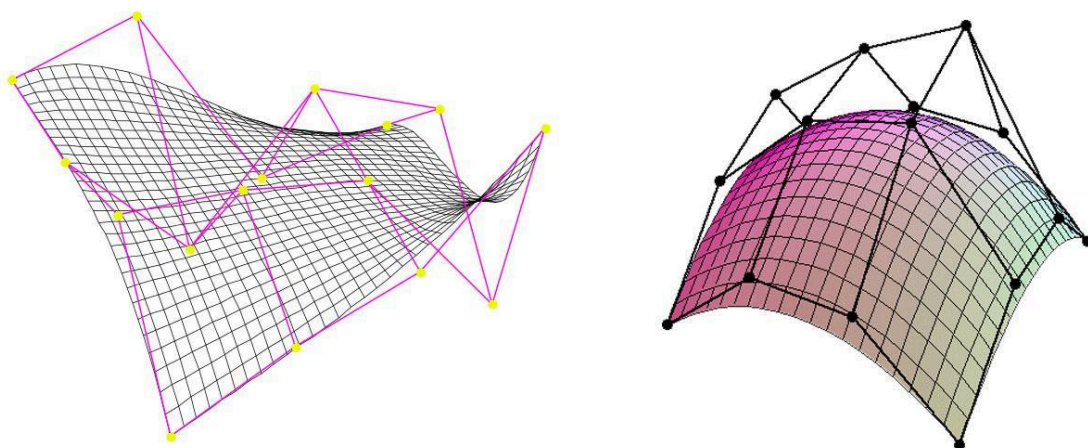


Figure 8 Objects created by NURBS (Roy, 2023).

Figure 7 shows the results of using NURBS with control points to create shapes.

Sculpting

3D sculpting allows artists to manipulate a model with tools that resemble working with clay. Artists can push, pull, pinch, or add additional mesh using the sculpting tools built into software. This method of modelling allows for the creation of complex shapes and textures especially while modelling organic objects like characters and trees. Before being used, however, sculpted models require re-topologizing which involves reducing the number of polygons used to make the shape and getting it ready for texturing. This step is necessary because the high polygon counts created by sculpting will slow down rendering when used in renders or video games. (Heginbotham, 2023).

Textures

To add features and realism to the surface of 3D models, textures are used. They are 2D images that can be applied to things to add patterns and colours or to replicate materials like fabric, metal, and wood. Additionally, textures can be imported from other sources or generated procedurally within the 3D software.

UV Maps

A UV Map is a 2D representations of the surface of a 3D model that is used to apply texture to it. UV Maps are created by 'unwrapping' a 3D object so that all its faces can be seen on a 2D image. These faces can be precisely moved around the image to assign specific parts of the image to them.

Unwrapping involves marking seams along edges of a model where the object will be cut to create the map (Denham, n.d.).

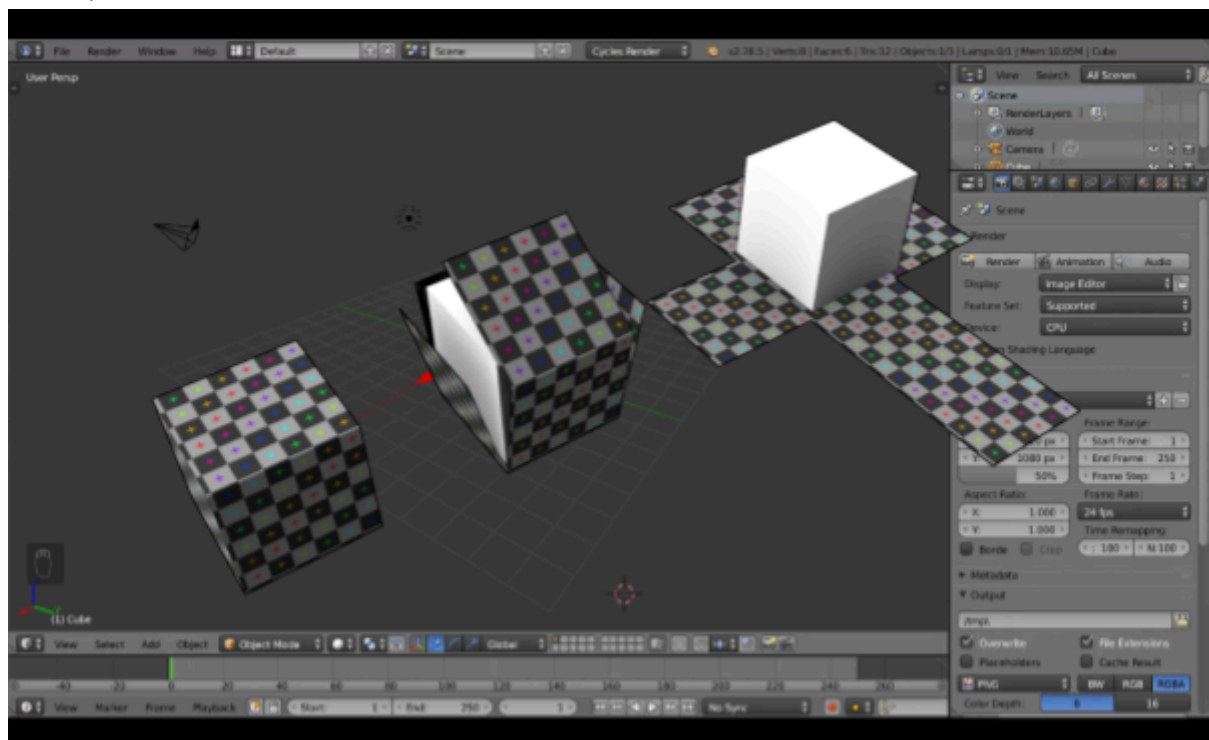


Figure SEQ Figure * ARABIC 9 UV Unwrapping CITATION Ble17 \ 2057 (Blender, 2017).

Figure 6 shows how UV Unwrapping works on a cube and how the faces are translated to 2D space.

PBR Texturing

Physically Based Rendering (PBR) textures mimic how real materials respond to light and how light interacts with various surfaces. PBR textures store surface and colour information within

two-dimensional images that can be applied to 3D objects (CGHERO, n.d.). These textures give precise reflection and bump information which can make a flat surface appear to have more depth and detail, so that models can have fewer polygons leading to decreased rendering times in the

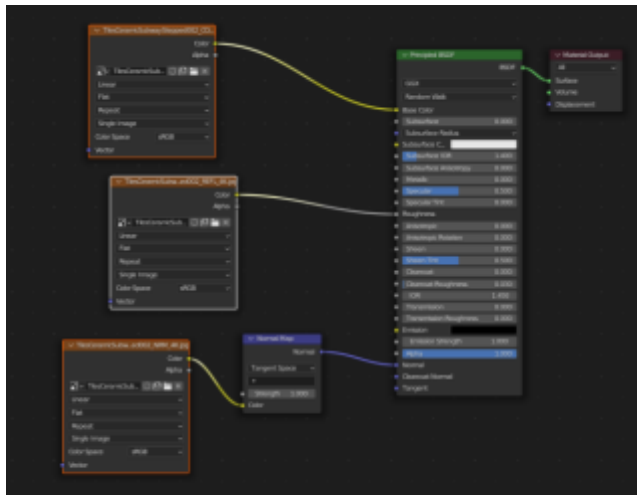


Figure SEQ Figure * ARABIC 11 PBR tile texture applied to a sphere.



Figure SEQ Figure * ARABIC 10 Result of PBR texturing.

viewport and final renders.

Figure 7 shows a simple node layout for using PBR textures using the Colour, Reflection and Normal images to create the result in Figure 8

PBR texture can be downloaded from websites such as [Poliigon](#) or [Textures.com](#) or created from scratch using software like [Substance Designer](#). PBR textures typically include diffuse, normal, specular, roughness/metallic, and ambient occlusion maps, which are combined in a material shader to create a realistic surface appearance.

Procedural Texturing

Blender allows users to generate intricate materials and textures for their 3D models using this program's node-based shading system, which is a strong tool. Users have the option to build a network of nodes that determine how light interacts with their models rather than depending on conventional texture maps and layers. With this technology, users may modify and fine-tune certain aspects of their materials in real-time, giving them more freedom and control over how 3D models look.

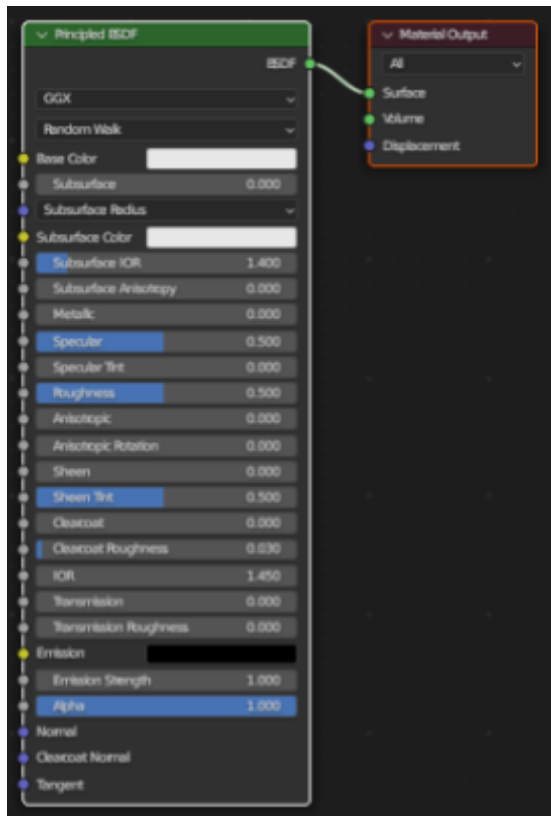


Figure SEQ Figure * ARABIC 12 Default blender shading node configuration.

Several nodes can be joined to form a material in the system. A separate characteristic of the material, such as colour, roughness, or transparency, is represented by each node. Complex materials may be created by manipulating and connecting nodes, and users can see changes in real time utilising Blender's viewport.

Figure 6 shows the standard node layout that is created when making a new material in Blender. The circles on the left of each box represent the inputs, and the right, the outputs.

The adaptability of Blender's node-based shading system is one of its main benefits. Users may simply adjust specific parameters to get the desired look since each element of the material is represented by a node in the material. Additionally, nodes may be merged and reused in many ways to produce new materials, speeding up and improving the efficiency of the material generation process. This method is also non-destructive as the output can be altered at any point by moving, removing, or adding nodes around or changing the values within them.

Physics

The ability to create realistic and dynamic animations is one of the key features of 3D software that relies on physics simulations. 3D artists may produce animations that precisely mimic the behaviour of things in the actual world by imitating real-world physics. This may be particularly crucial in fields like movies and video games, where realism is essential to a project's success.

The Bullet Physics library is a well-liked physics simulation tool in 3D applications. Blender and Maya both employ the open-source Bullet physics engine, which is a component of many 3D software programmes. Rigid body dynamics, soft body dynamics, and cloth simulations are just a few of the physical effects that Bullet is capable of simulating.

The NVIDIA PhysX engine is another popular physics engine utilised in 3D applications. Numerous applications, including video games and visual effects for movies, use PhysX. Numerous physical effects, such as rigid body dynamics, soft body dynamics, and fluid simulations, may be simulated using the engine.

Physics simulations are also used in a variety of fields apart from film, television, and gaming, including engineering and architecture. Engineers and architects may optimise their designs and find possible problems before construction starts by using physics simulations to model the behaviour of buildings and materials under various situations.

Rigid Body

Rigid body simulations involve simulating the physics of solid objects that do not deform. In a rigid body simulation, physical characteristics like mass, friction, and gravity are assigned to objects, and those objects simulated using the provided parameters. This feature is often used to simulate falling objects, collisions, and explosions (Blender, 2022).

Active rigid bodies were applied to the objects that make up the top layer. The mass was changed using the built in 'Calculate Mass' feature and an appropriate density was input. This uses the volume of the object and the supplied density to calculate its mass and can be done to all selected objects at once. A margin of 0m was important so that the objects must touch to collide, meaning that the simulations would look more realistic.

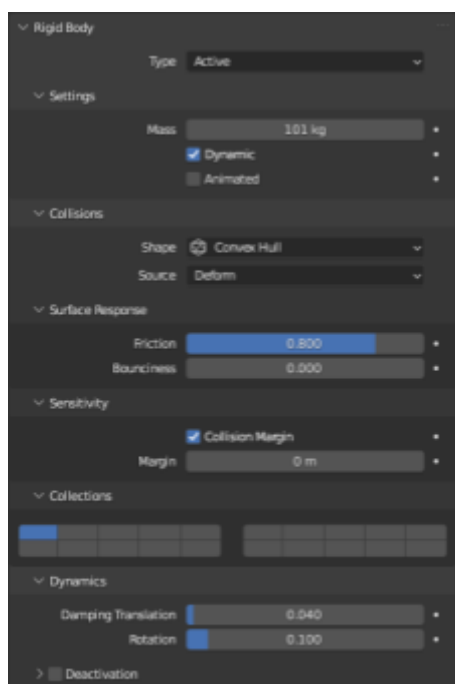


Figure SEQ Figure * ARABIC 13 Active Rigid Body settings.

The lower layer uses a passive type of rigid body so that it itself is not affected by gravity but still collides with other rigid bodies. The animated box is checked so that the rigid body updates with the keyframe animation specified in the animating section. Its margin is also set to 0m.

This object also has a collision physics modifier so that it will interact with emitted particles. The damping and friction parameters were increased so that the objects would absorb the force of particles, slowing them down. These settings can be seen in Figure 41.

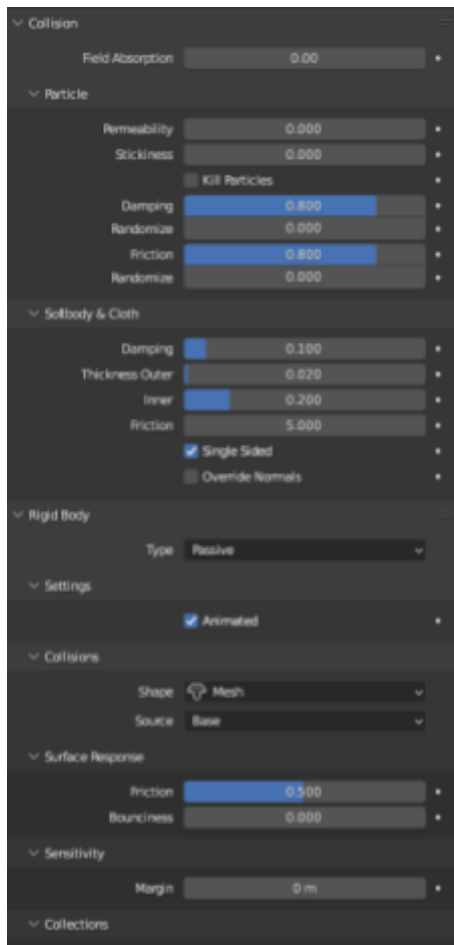


Figure SEQ Figure * ARABIC 14 Lower layer physics settings.

Soft Body

Soft body simulations model the physics of flexible materials like skin, rubber, and fabric. They are often utilised in 3D software to produce realistic animations. In a simulation of a soft body, objects are given physical characteristics like elasticity, stiffness, and damping, and are then let to interact with one another in accordance with those characteristics. This makes it possible to simulate animations like waving flags, bouncing balls, and character movement (Blender, 2023).

Cloth

Cloth simulations have many of the same features as soft body simulations, but they are not compressible. This means that each vertex is connected by edges with fixed length rather than being variable.

The pylon in the project uses cloth physics for its animation rather than soft body as it is simulating a metal object with fixed length rods. The settings were adjusted to add rigidity. The result was the structure collapsing from the bottom up but without any individual edges bending, as if it had been shaken by an earthquake and failed to stand.

Fluid

In a 3D setting, fluid simulations simulate the mechanics of liquids and gases. They are frequently utilised in 3D software to provide lifelike animations of fluid effects like water, smoke, and others. Physical parameters like viscosity and surface tension are assigned to objects in software by the user

and can be simulated to animate things like the movement of clouds and the flow of water over rocks or the rise of smoke from a fire (Blender, 2022).

Fluid physics were used to create dust clouds. This involved creating a smoke domain object that would contain the whole scene and applying fluid to emitter objects on either side of the crack. The domain was given a custom shader to make it look more like dust and the density was adjusted to a reasonable value.

Particles

Particle systems create instances of objects that have unique behaviours such as mass, velocity, and lifespan. These instances are emitted from a mesh or point and are subject to various forces and constraints like gravity, wind, and collisions that affect their movement.

Hair particles also exist which scatters the instanced objects on the surface of the mesh and can be used to simulate systems like hair or grass.

Rocks were created using the add-on 'Extra Objects' (Blender, 2023). This add-on creates objects based on parameters adjusted by the user. These objects were placed in a collection (Blender, 2023) and set as the source for the particle system with randomised scale, as can be seen in Figure 40. The objects were given mass based on their volume and physics were activated so that they would interact with other objects that had collision physics applied in a convincing way.

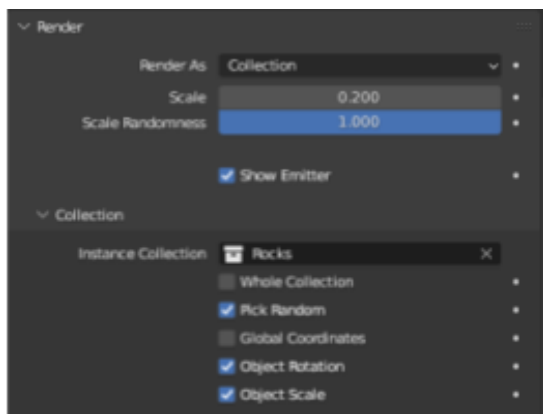


Figure SEQ Figure * ARABIC 15 Particle render settings.

Force Fields

Force Fields are used to influence the movement of other objects that have physics applied to them. Examples of force fields are magnetic fields, vortex fields or wind. Each force field type has its own properties and affects objects differently. They usually have parameters such as strength, flow and falloff which are used when simulating movements. Force Fields can be restricted to affecting only certain objects so that users can get their desired result (Blender, 2023).

Wind was added to the scene using an object with a wind force field applied (Blender, 2023). This affects the fluid and particle systems. This makes the dust emerge from the ground but does not greatly affect the rocks as their weight is greater than the upward force.

Figure 16 shows the configuration of the physics in the scene.

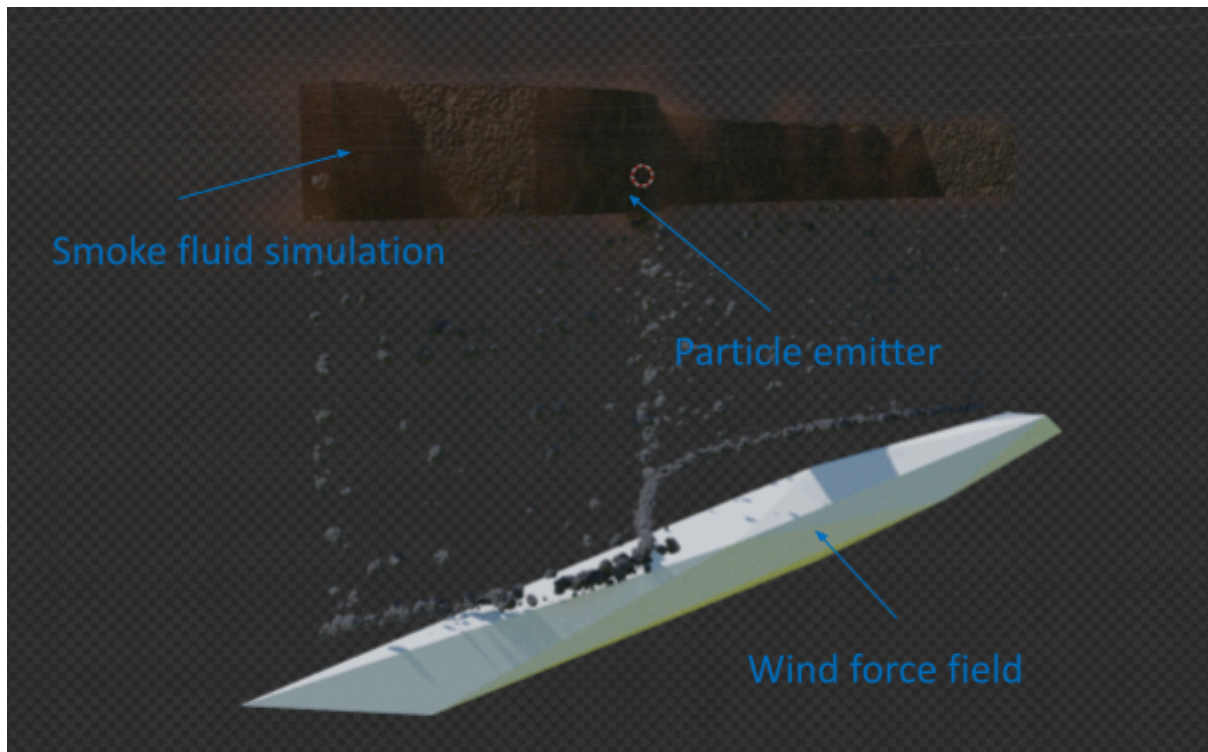


Figure SEQ Figure * ARABIC 16 Particle, Fluid and Force Field physics.

Baking

Baking is an important step in the process of simulating physics. It entails calculating and documenting how objects in a scene move and behave as they engage with the simulated environment. These simulations may be played back in real-time after being baked, providing more precise and effective previews of the finished animation.

Baking improves the performance of the software as it no longer has to calculate the physics every time the simulation is played. Baking also means that the animations will not change if objects in the scene are moved around or removed. The bakes are saved in the cache of the software and can be transferred to other files or reloaded to use the simulation more than once (Blender, 2023).

The project physics simulations were baked together using the 'Bake All Dynamics' option in Blender that ensures that the separate physics systems involved in the scene bake together and therefore correctly calculates their movement. This improved processing times while working on the project as they did not have to be calculated repeatedly.

Animation

Animation is the process of adjusting object properties like position, rotation, or scale over time to give the illusion of movement.

Animations are typically created by setting keyframes at specific points in time. To produce smooth animation, the programme interpolates the values between keyframes, which are used to specify object properties at a certain point in time (Blender, 2023).

Rigging, which is the process of building a digital skeleton for a figure or object, is another aspect of animating. The movement and distortion of the item or character are then controlled by the rig. The gear may be adjusted by animators to produce a variety of motions, including walking, running, leaping, and changing face expressions.

The only objects in the scene that were hand animated were the two objects that make up the lower layer. Their position was set with keyframes at frame 1 with both objects touching, and again at frame 570, near the end, where both objects were moved in opposite directions along the global y-axis. This allowed enough space for some of the fragmented mesh of the layer above to fall through the gap.

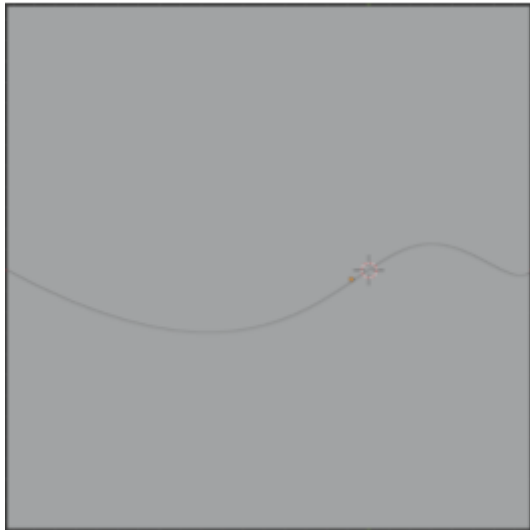


Figure SEQ Figure * ARABIC 18 Frame 1.

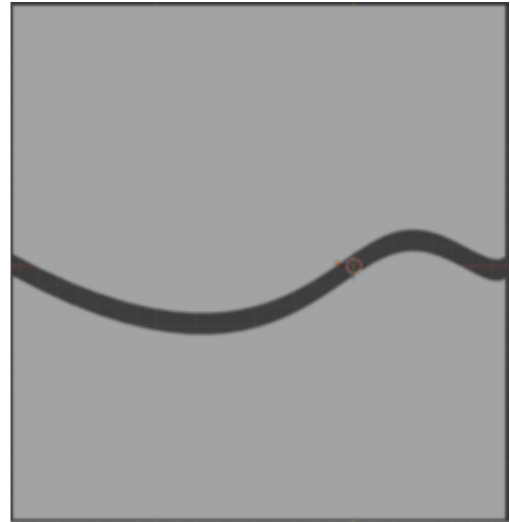
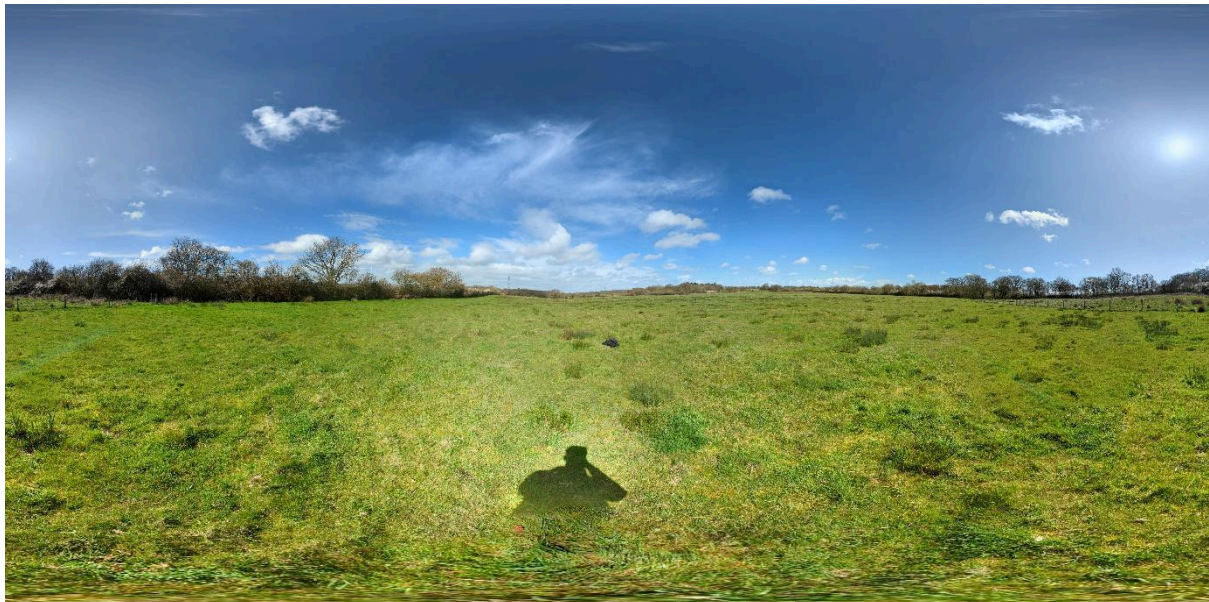


Figure SEQ Figure * ARABIC 18 Frame 570.

Lighting

High Dynamic Range Images

A unique kind of image called an HDRI (High Dynamic Range Image) is used in 3D graphics and visual effects to imitate realistic lighting. It allows for more precise lighting information to be used in a scene because it contains a far larger range of brightness and colour values than a typical image.



HDRI images are frequently produced by taking several photos of the same scene at various exposures and merging them to get a single image that fully captures the lighting data. Using HDRI photos, incredibly realistic lighting conditions, reflections, and shadows can be produced in 3D

software. They are frequently employed in compositing to synchronise the lighting of 3D components with real-world video (Segasby, 2022).

At the time of filming, a photo sphere image was taken which captures the whole environment by taking multiple images from all angles, like a panoramic image but in 360 degrees. This image would later be used as an environment texture within Blender so that the lighting of the scene could be matched to the outdoor lighting in the footage.

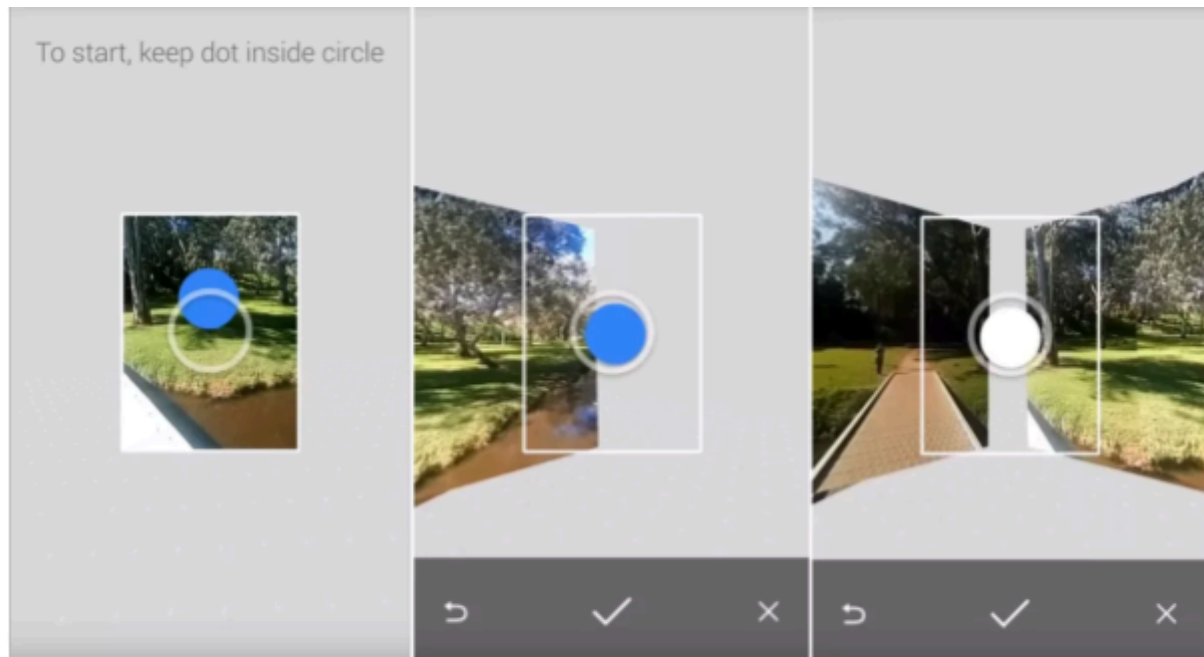


Figure SEQ Figure * ARABIC 19 Screenshots of Android photosphere camera mode (Karch, 2021)

Composite

Compositing is the step in the visual effects workflow where the artist combines visual elements such as live action footage, 3D generated imagery and 2D imagery into a single, cohesive shot.

Different software handles compositing in different ways.

The industry standard software, Nuke, has an advanced range of features that means it is preferred by large studios. Nuke uses node-based editing to create intricate projects, which means you connect different nodes to each other that alter the image and is considered one of the best options for professional use.

Adobe After Effects uses layer-based compositing in its software which means that layers are stacked on top of each other with the lower layers appearing under the top layers. It is much easier to learn as a beginner but still offers a range of tools that allow the artist to fine tune their composites. It is used more frequently by individuals and smaller studios (Trussell & 9to5Software, 2022).

Alpha, Masks and Mattes

Alpha channels are used in compositing to distinguish between the foreground and background of an image or video. An image's transparency is represented by the alpha channel, where a value of 0 denotes total transparency and a value of 1 denotes total opaqueness. Masks that can be used to isolate areas of an image or video are frequently created using alpha channels.

The same tools found in compositing software, such as roto brushes and keying tools, can be used to build and modify alpha channels. They can also be used to stack various alpha channels on top of one another to build complicated composites. For instance, the green backdrop can be keyed out and replaced with another background using the alpha channel of a person in front of a green screen (Adobe, 2022). Tracking can be used in conjunction with the alpha channel to isolate a subject from the background of a video.

Colour Correction

Colour Correction is the process of adjusting the colours in a video to improve its appearance. This involves adjusting brightness, contrast, saturation, hue, and other aspects of colour. The aim of colour correcting is to balance footage to look consistent and balanced with the rest of the project. This can be useful if some of the footage has undesirable lighting but is otherwise perfect (MasterClass, 2021).

Colour correction is also used when introducing visual effects to a shot as it is unlikely that the visual effects match the colours of the original footage immediately. The colours can be adjusted so that the added visual effects don't look out of place and makes the shot more seamless and realistic.

Colour Grading

Once Colour Correction is complete, Colour Grading can be used creatively to achieve a desired mood or atmosphere for a shot. At this stage, the vision of the filmmakers can be shown through the colours used. Different colour profiles can be applied to stylize the shot to invoke different feelings. "For example, *American Beauty* (1999) uses more red in dramatic moments to represent passion, anger, and power, where *Maleficent* (2014) uses green to convey danger, corruption, and darkness." (MasterClass, 2021).

Development Process

This section will explain the process the project took to reach its completed form, along with a log of the changes made to it over its course. It will cover the methods used in the production of the visual effects shot and any relevant technical details.

Versions

Version 0

Using a [YouTube tutorial by Blender Made Easy](#) this image was created. The result is a plane where the geometry along the defined crack has been moved down. It uses a complicated node setup that can be seen in Figure 13. This effect worked well for still images but would not adapt well to objects moving as the crack could not be animated.



Figure SEQ Figure 1* ARABIC 20 Version 0.

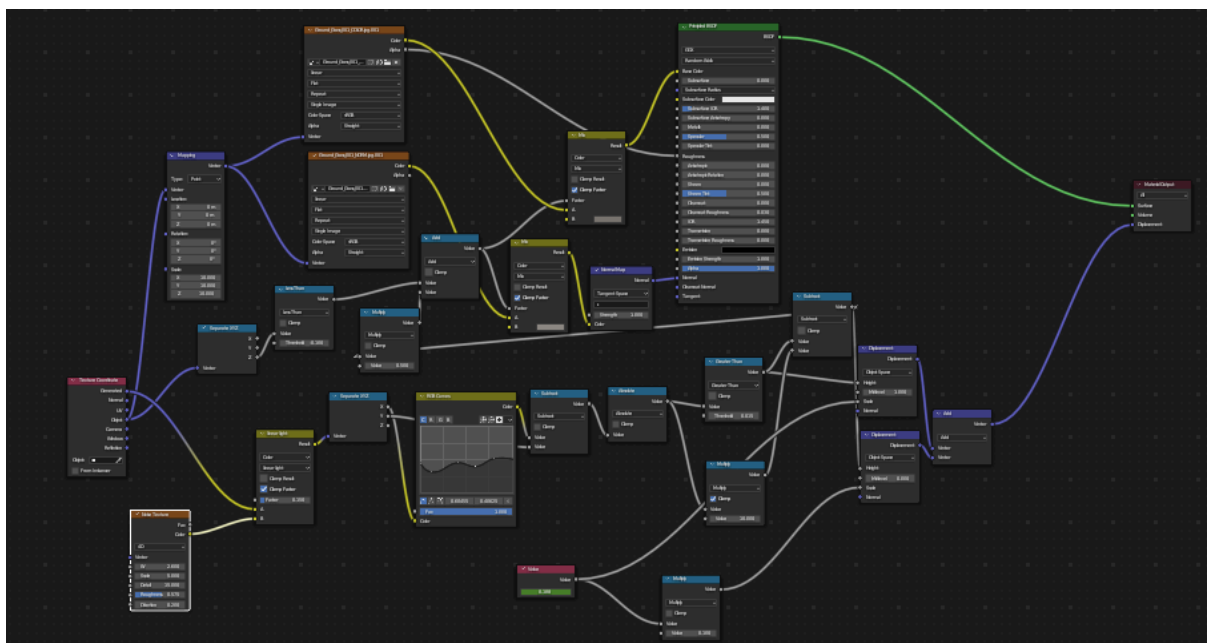


Figure 21 Node setup for Version 0.

Version 1

This version was the first iteration of using the Cell Fracture addon to create individual pieces of ground. Rigid Body physics were applied to them so that they would behave realistically. This approach was effective but had some issues with certain objects not colliding properly, causing pieces to fly into the air.

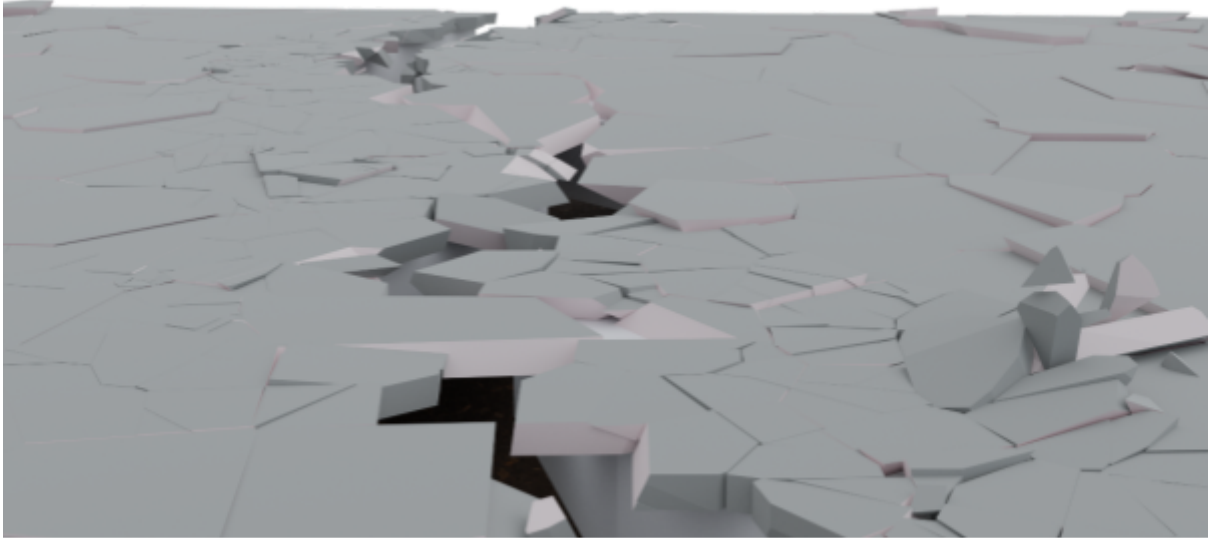


Figure SEQ Figure * ARABIC 22 Version 1.

Version 2

In this version, grass and debris were added using Particle Systems. The grass used a hair system, and the debris used an emitter system. Elements of this version were successful, especially the falling rocks, but they interacted strangely with the ground, bouncing up rather than hitting the ground.

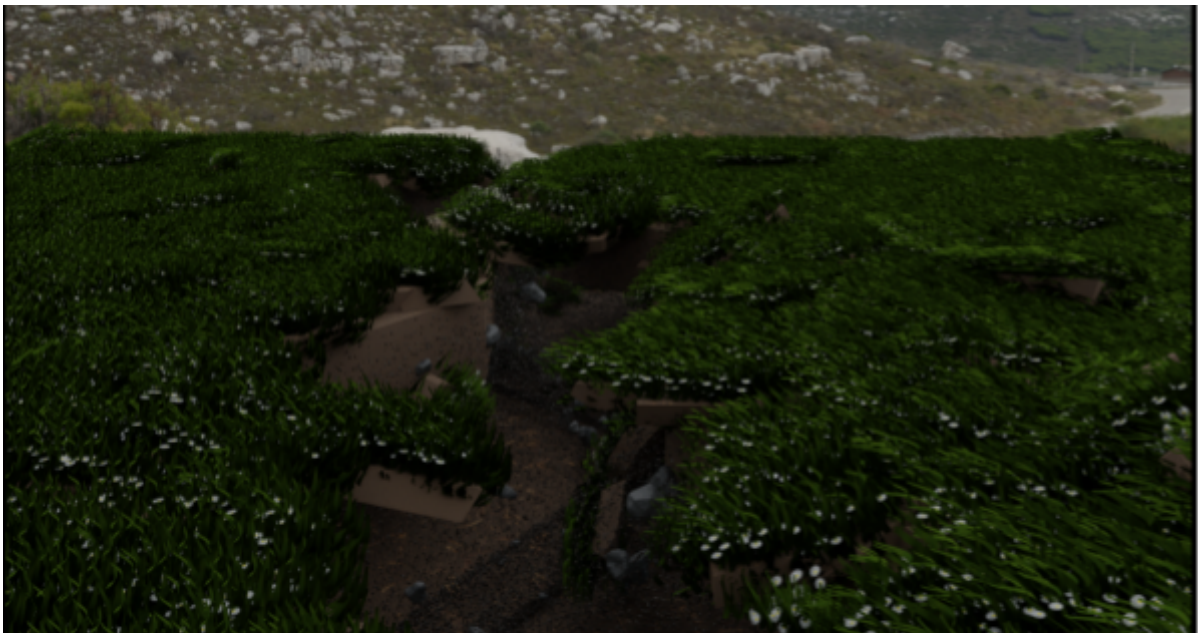


Figure SEQ Figure * ARABIC 23 Version 2.

Version 3

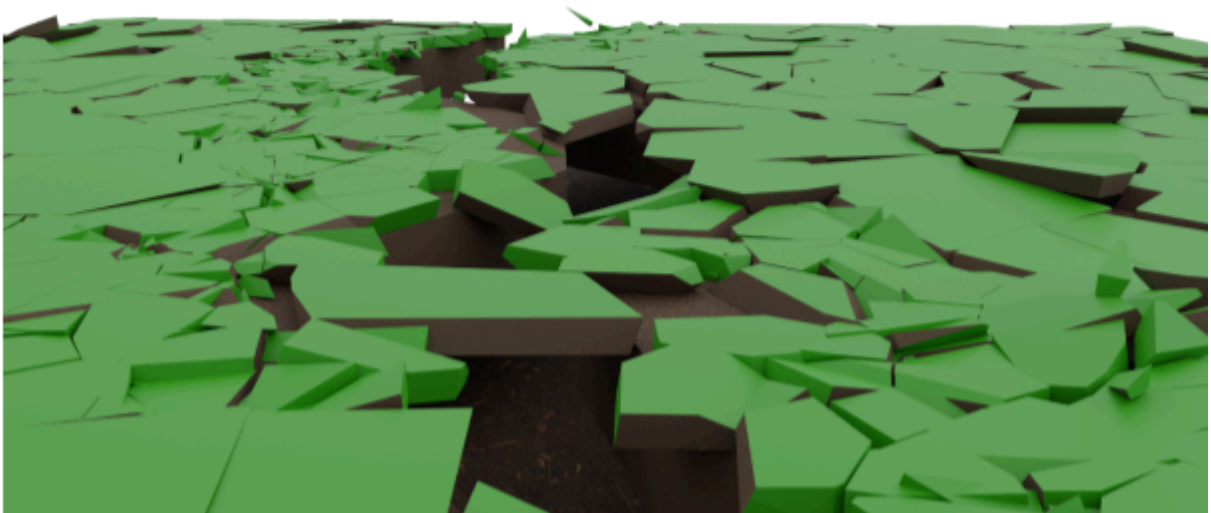
Dust clouds were added in this version using a smoke simulation. The parameters were not set properly and as a result, the smoke was too dense. This obscured the view of the scene but would look good if the input values were adjusted.



*Figure SEQ Figure * ARABIC 24 Version 3.*

Version 4

This version attempted to solve the issue of the ground not looking realistic by adding a green texture to the top surfaces of the ground that would be keyed out in After Effects with the Keylight effect. Using the eyedropper, the green was selected, and an image of a field was placed below the rendered layer to create the result in Figure 18. This did not work because the image did not move with the pieces so the texture on each fragment was changing as the crack grew.



*Figure SEQ Figure * ARABIC 25 Version 4 Blender render.*



Figure 26 Version 4 After Effects result.

Version 5

In version 5, a similar strategy was used but using an alpha matte instead of green screen. The render would act as the transparency data for the field image. This did not change the result and the ground texture was still not applied correctly.

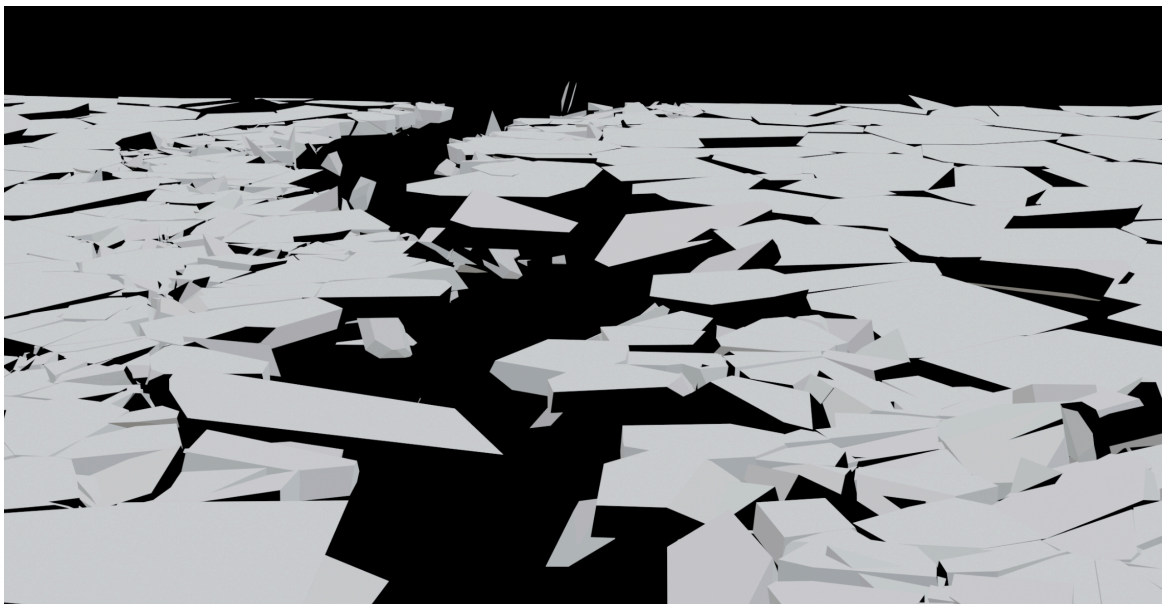


Figure 27 Version 5.

Version 6

In this version, projection mapping was used to apply the image directly to the ground, meaning that the individual pieces of ground would have a corresponding section of the image attached to them. This was effective because the ground now looked like it was splitting. The texture did, however, look very flat.

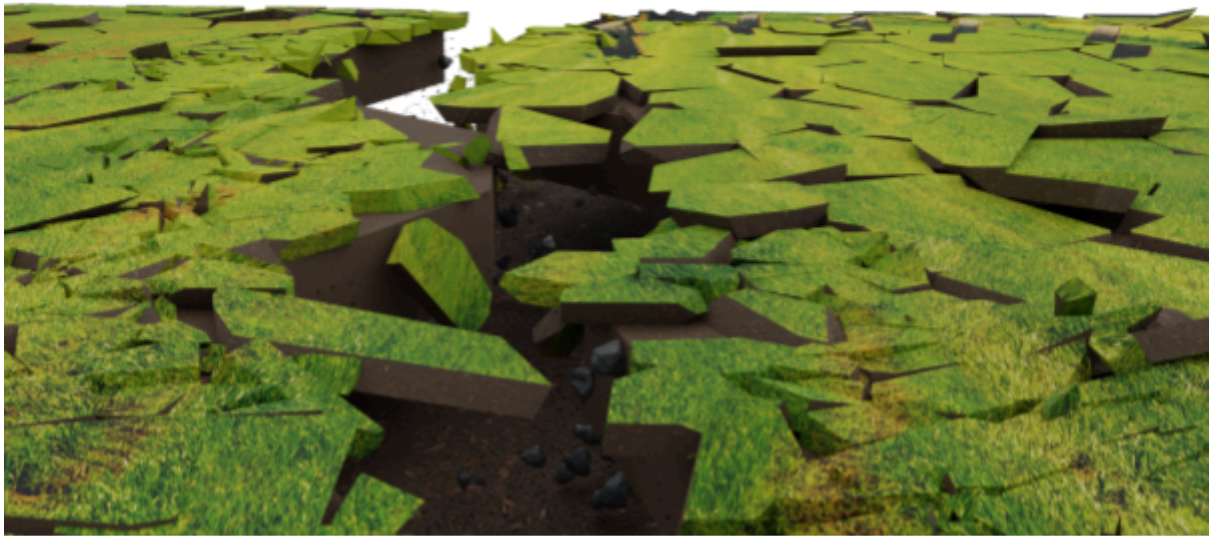


Figure SEQ Figure * ARABIC 28 Version 6.

Version 7

In version 7, the grass particles were recreated so that they better matched the project footage. It was accomplished using nodes instead of the built in particle system. Figure 22 shows the node setup for the result in Figure 21. The result is a fairly realistic field of grass but when compared to the footage, the grass patterns appear too consistent and uniform.

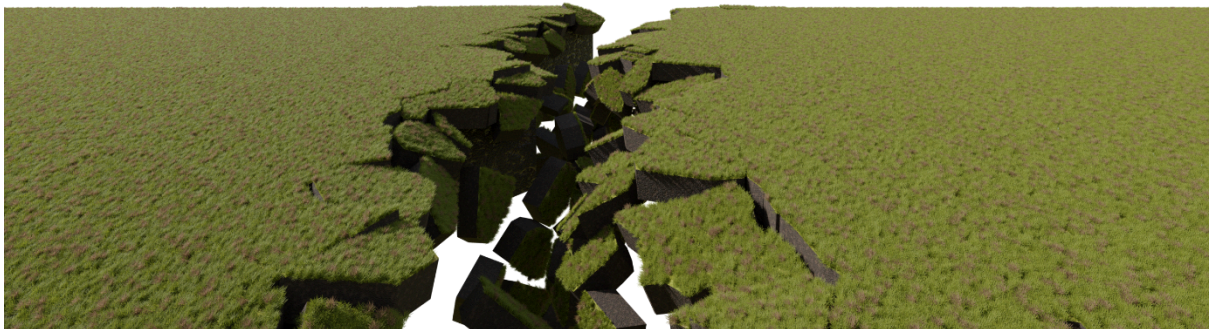


Figure 29 Version 7.

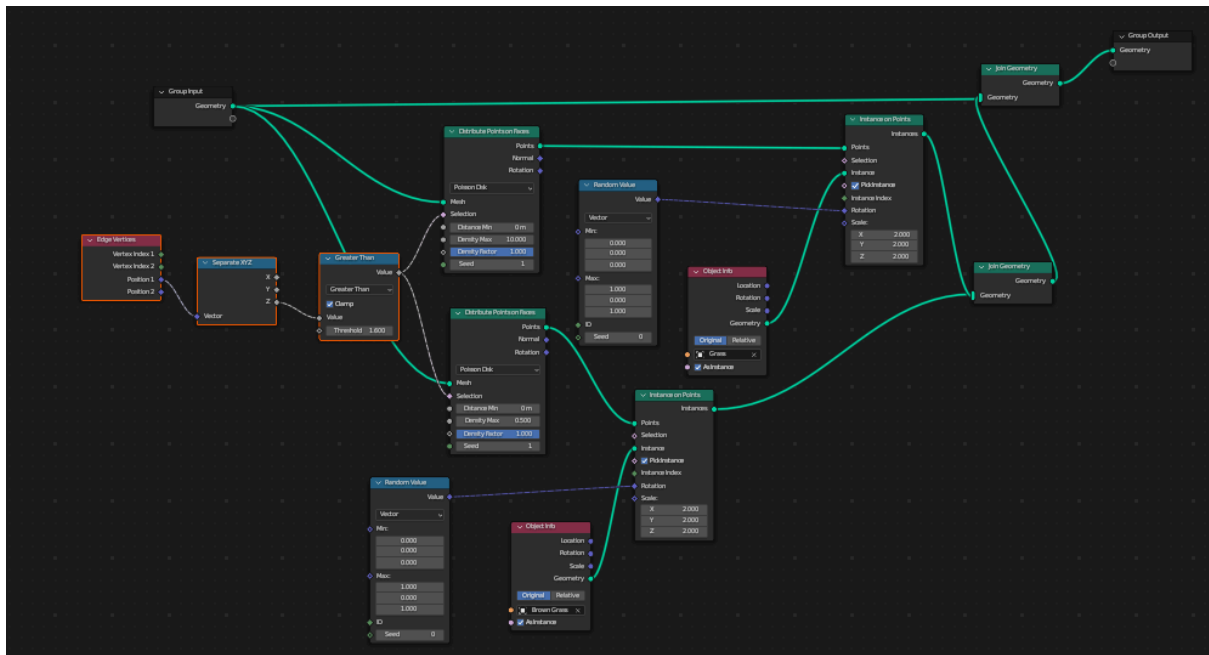


Figure 30 Version 7 Geometry Nodes.

Version 8

This version combines many of the techniques of the previous versions by using projection to texture the ground, particle systems for the rubble and smoke simulations for the dust but were tweaked to enhance the realism of the result. 3D tracking of the footage was used so that the movement of the Blender camera would follow that of the camera in the footage. The movement of the camera increased the immersion of the shot.



Figure 31 Version 8.

Pre-production

Prior to any footage being filmed or animation being created, the pre-production phase must take place. This is where planning, including storyboarding, scriptwriting, casting, location scouting, budgeting, and scheduling happens. Pre-production aims to guarantee that the project is carefully thought out and organised, as well as to reduce the possibility of making expensive mistakes during production.

Planning

Visual Effects Project Timeline

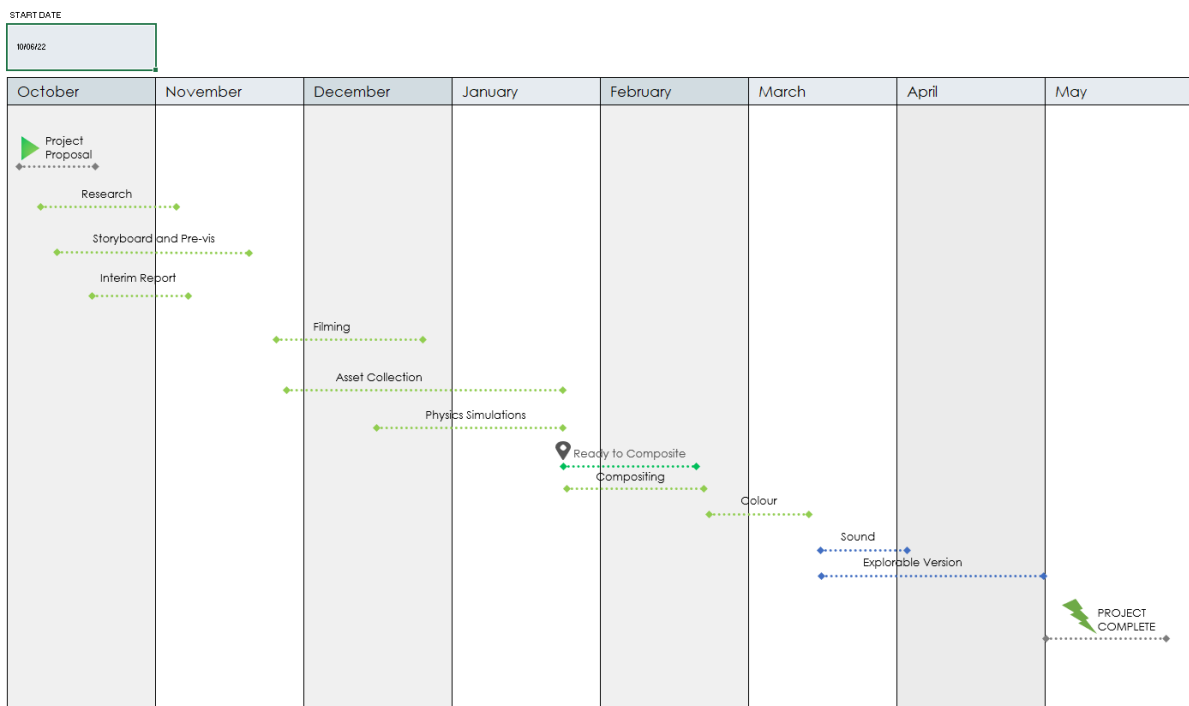


Figure 32 Project development timeline.

	Monday 10 Oct	Tuesday 11 Oct	Wednesday 12 Oct	Thursday 13 Oct	Friday 14 Oct
08:00					
08:30					
09:00	Human-Computer Interaction (Lecture 1) Arts A A02			Visual Effects (Laboratory 1) Chichester 1 CI204/5	
09:30					
10:00					
10:30					
11:00					
11:30	Project work				
12:00	Project work			Visual Effects (Lecture 1) Richmond AS03	
12:30	Project work				
13:00					
13:30			Project work		
14:00		Individual Project (Lecture 1) Chichester 1 CHIC 1-LT			
14:30					
15:00	Programming for 3D (Laboratory 1) Online See Canvas			Project work	Programming for 3D (Laboratory 1) Online See Canvas
15:30					
16:00					
16:30					
17:00					
17:30					
18:00					
18:30					
19:00				Human-Computer Interaction (Seminar 1) Arts A A02	
19:30					
20:00					
20:30					
21:00					
21:30					

Figure SEQ Figure * ARABIC 33 Working schedule.

Storyboard

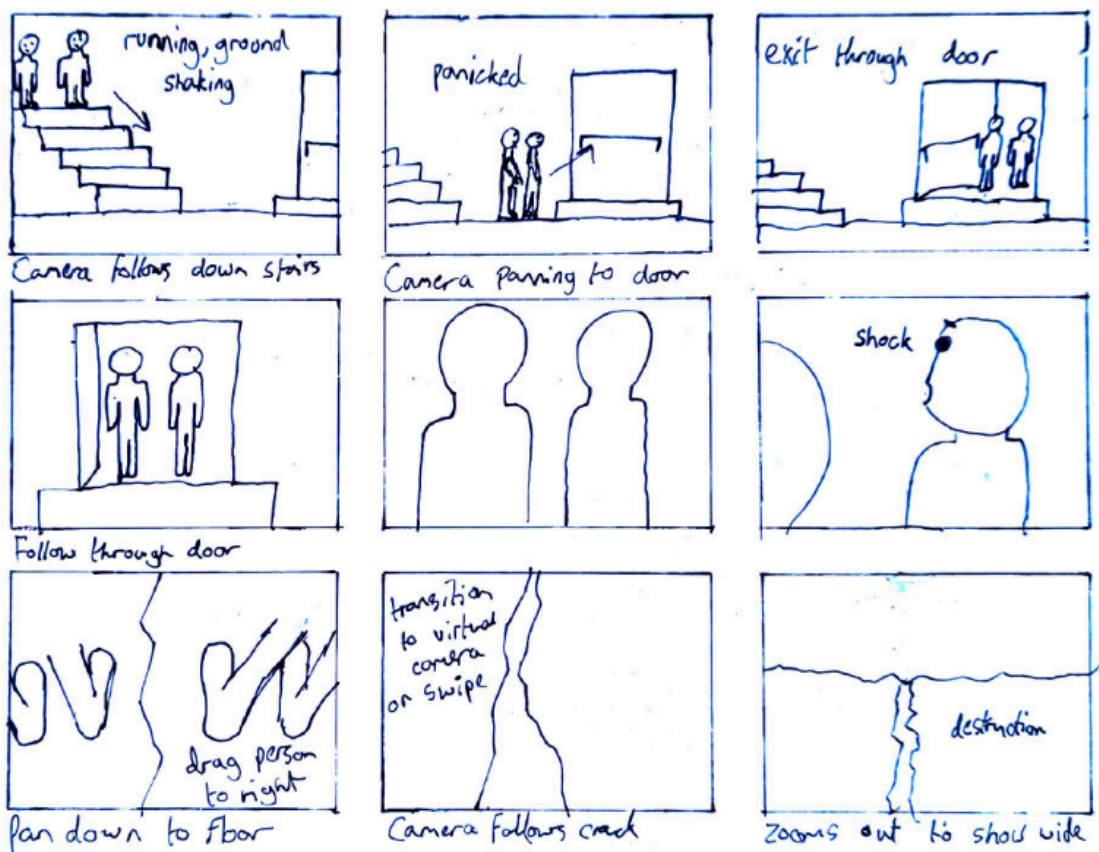


Figure 34 Initial storyboard for the shot.

The actual shot may differ from the storyboard as the project develops.

Pre-vis

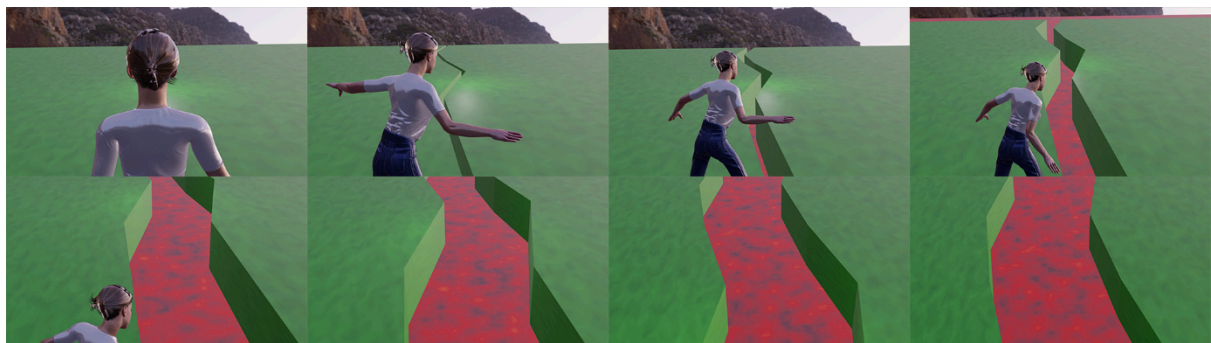


Figure 35 Pre-visualisation of the shot.

Production

Filming

Post-production

Modelling

The crack effect is created using a top extruded plane layer divided into pieces, mesh below that separates to allow the pieces above to fall and a container for the scene which keeps all the elements affected by physics together and controls the simulation.

The top layer was created using an add-on for Blender called Cell Fracture (Blender, 2023) which is included in Blender but needs to be activated in the preferences. It takes a mesh object and divides it

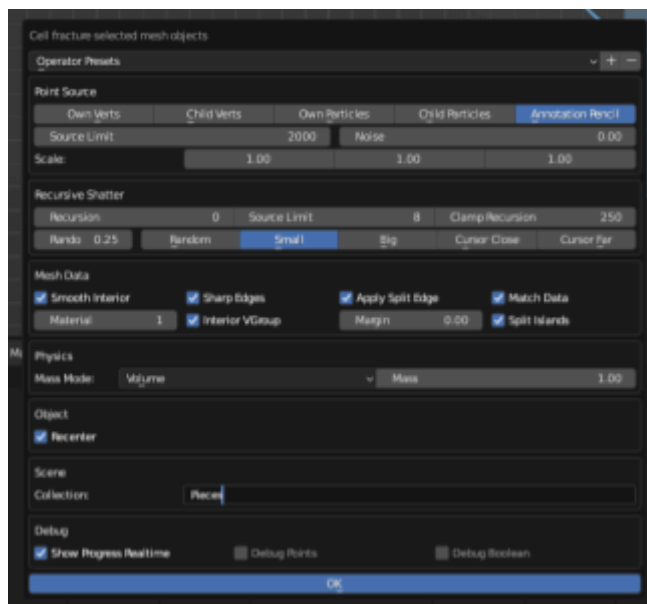


Figure SEQ Figure * ARABIC 37 Cell Fracture Settings Menu.

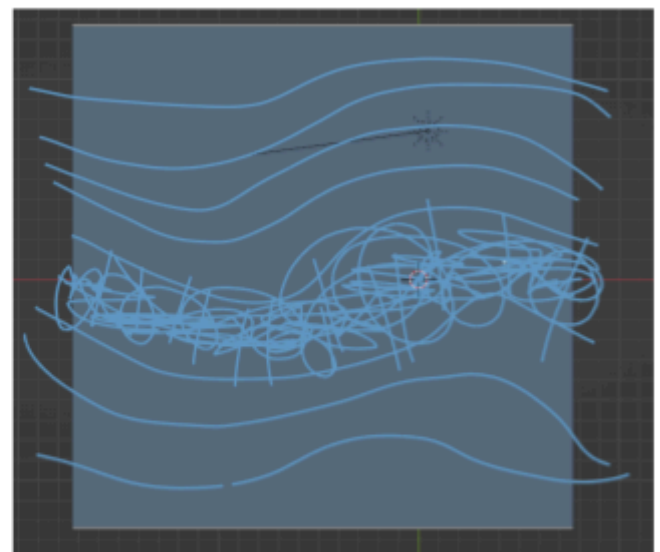
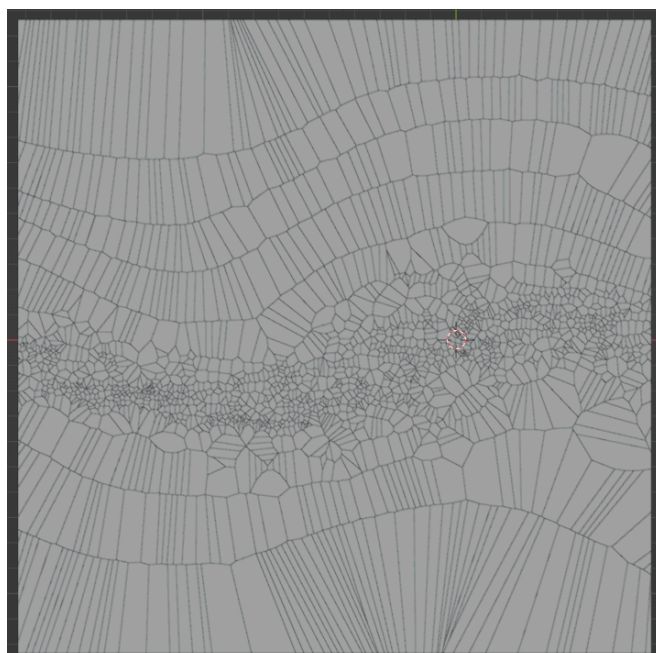


Figure SEQ Figure * ARABIC 37 Annotation Pencil to determine mesh shape.

into procedurally generated pieces, depending on the user settings. It then creates separate objects from these pieces that can have physics applied to them individually.

In the settings for Cell Fracture, Blender's annotation pencil option is used so that the shape of the crack can be defined manually. Fewer lines are drawn further from the crack and more lines are drawn closer so that the program creates smaller objects near the crack. These settings were applied, and the result is shown below.



The bottom layer of ground was created by cutting through a cube using a curve object and the Knife Project tool. This created an edge along the cube and the shape was then separated into two objects, making sure to fill any holes in the mesh.



*Figure SEQ Figure * ARABIC 39 Lower layer split into two objects.*

The pylon in the background of the shot was modelled by extruding a cube and using the 'poke faces' option in Blender. This creates a new vertex in the centre of the face and connects it to the corners with edges, making a shape that looks like the supports of a pylon (Blender, 2023). A wireframe modifier was then applied to the object which takes the mesh and turns all edges into polygons with four sides, so that it looked like the object was made up of many beams (Blender, 2023). Next, the cable connectors were modelled out of cylinders with several loops around the circumference. Every other loop was scaled in the x and y axes to give it texture.

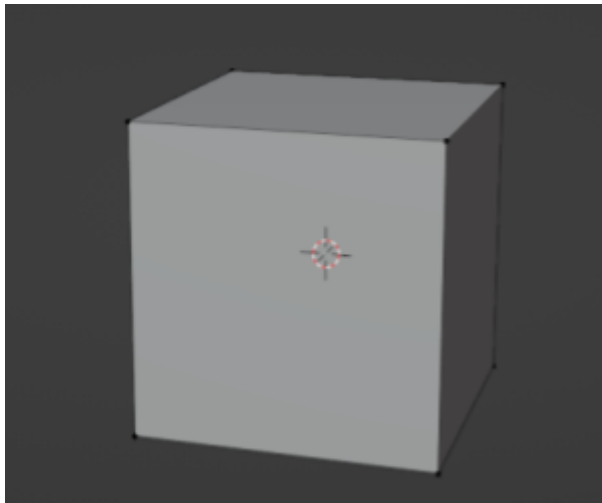


Figure SEQ Figure * ARABIC 43 Basic cube geometry.

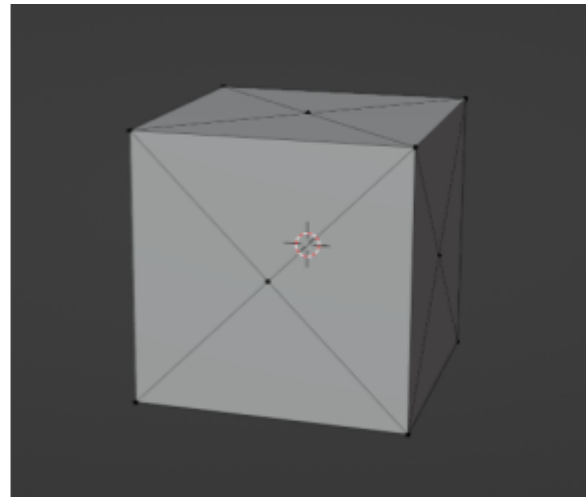


Figure SEQ Figure * ARABIC 43 Cube after Poke Faces is used.

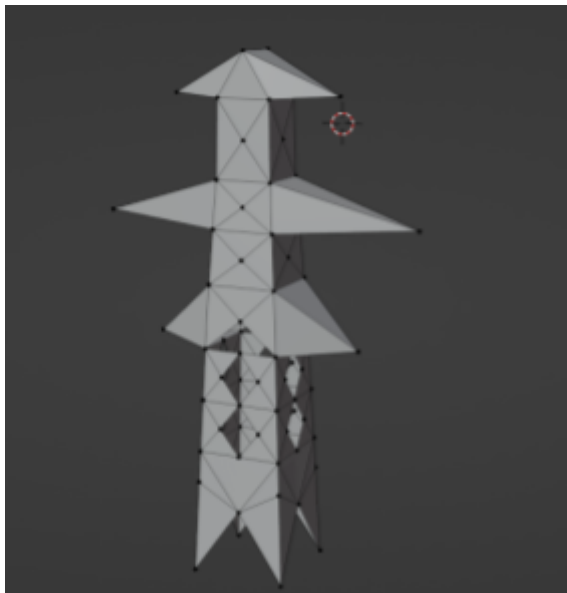


Figure SEQ Figure * ARABIC 41 Pylon model before Wireframe modifier.

Animating

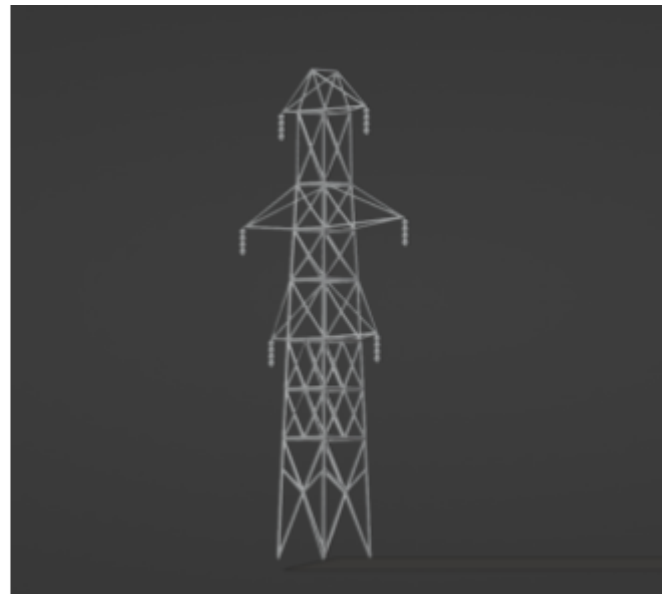


Figure SEQ Figure * ARABIC 41 Pylon model after Wireframe modifier.

Applying Physics

Motion Tracking

Texturing

The shading and texturing for this project was accomplished using Blender's node-based system.

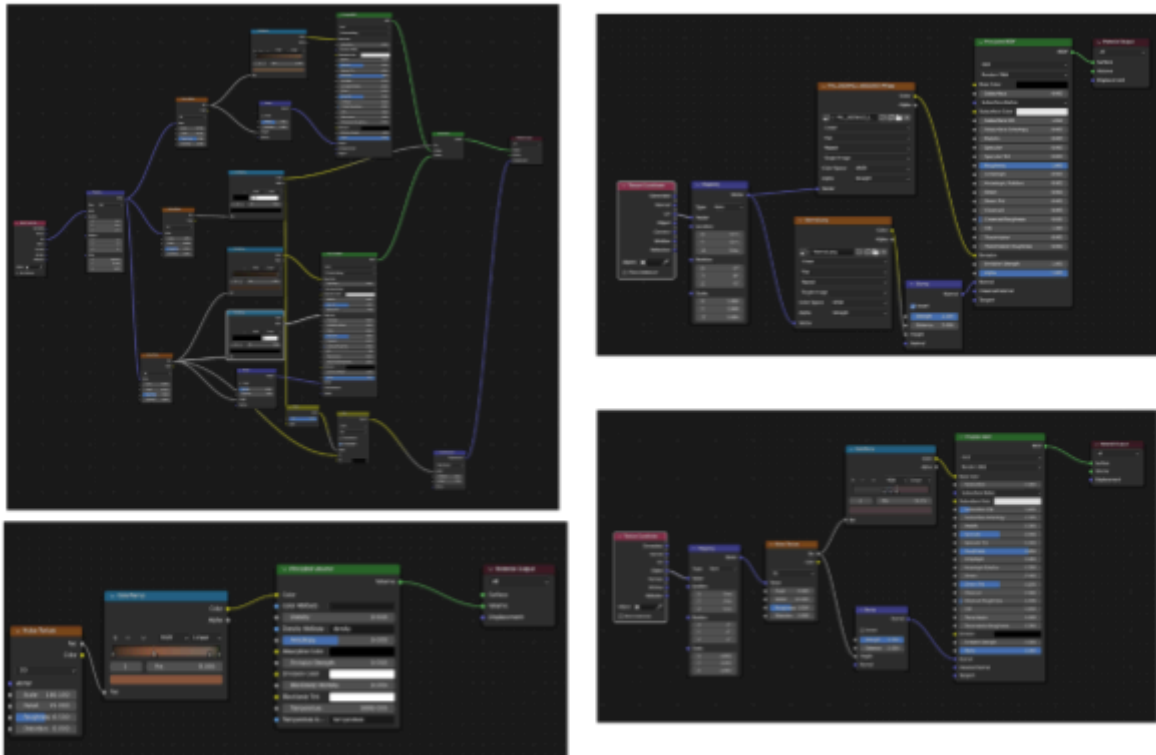


Figure SEQ Figure * ARABIC 44 Node setups for the materials in the scene.

The combination of Noise Textures and ColorRamp nodes were used frequently to apply colour to objects. Multiple colours were assigned to the ColorRamp and the Noise Texture created a map that decides where those colours should be seen. These nodes are useful to create variety in the colour so that objects look less uniform. The same Noise Texture is applied to a Bump node to emulate height difference on the objects which cast shadows.

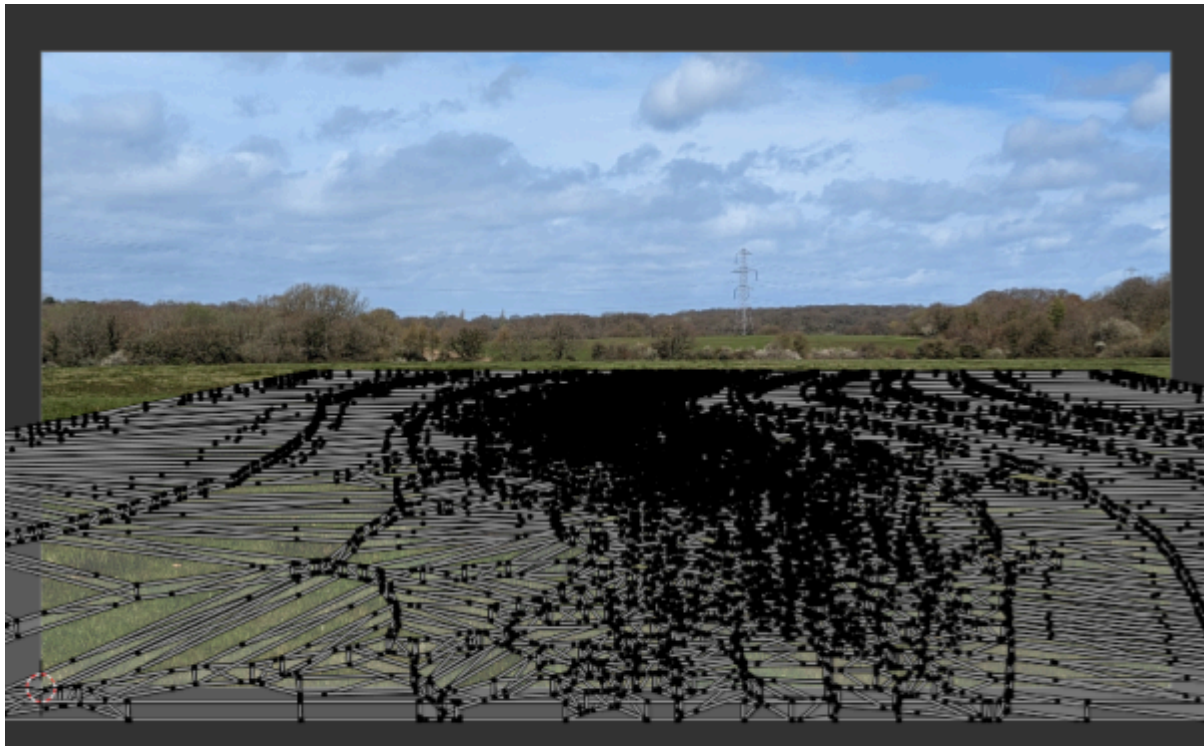


Figure SEQ Figure 1 ARABIC 45 UV map for ground made by using Project from View.*

For the top layer, a UV map was created using the 'Project from View' option while the view was aligned to the Camera. This meant that the image texture would match the perspective of the camera. Each face was assigned a portion of the image that was taken from the first frame of the footage.

Compositing

The main Blender file was rendered out as a sequence of PNG images at 4K 30fps. The PNG file format was used so that the images would be transparent on the pixels where no information was present, allowing it to be composited without needing to mask out areas. The Pylon file was rendered out with the same settings.

Once the render was complete, the next step was to composite the elements together in After Effects. Firstly, the sequence was imported and dropped into the timeline of a new composition, along with the original footage. These elements lined up well, but the edge of the render was hard, so a Roughen Edges effect was added to soften it.

A crack that goes through the background of the image was added using a shape layer drawn with the pen tool that has keyframes on its stroke width that increase over the course of the clip so that it appears as if it is growing.

The pylon in the original footage was removed using the clone stamp tool and an expression to track it to the footage using a motion tracker on the tip of the pylon. Figure 47 shows the result of using the clone stamp. Once the original pylon was removed, the rendered pylon animation could be added to the same position, using the same tracker as the clone stamp. A mask was applied to the pylon layer so that it was not seen in front of the foreground and looked like it was falling into the ground. The pylon was blurred so that it didn't look as sharp in the background.

Behind the gap in the background, a blue solid the same colour as the sky was added to fill the gap as well as a solid that uses the same dirt texture as the sides of the crack pieces. The dirt solid was given a roughen edges effect with the roughen colour option set to green to look like grass.

Particle systems were added on each side of the gap that would create particles coming from the treeline. They are tracked to follow each side of the gap as it grows and keep emitting particles from the correct location. The dirt texture was also applied to the particles.

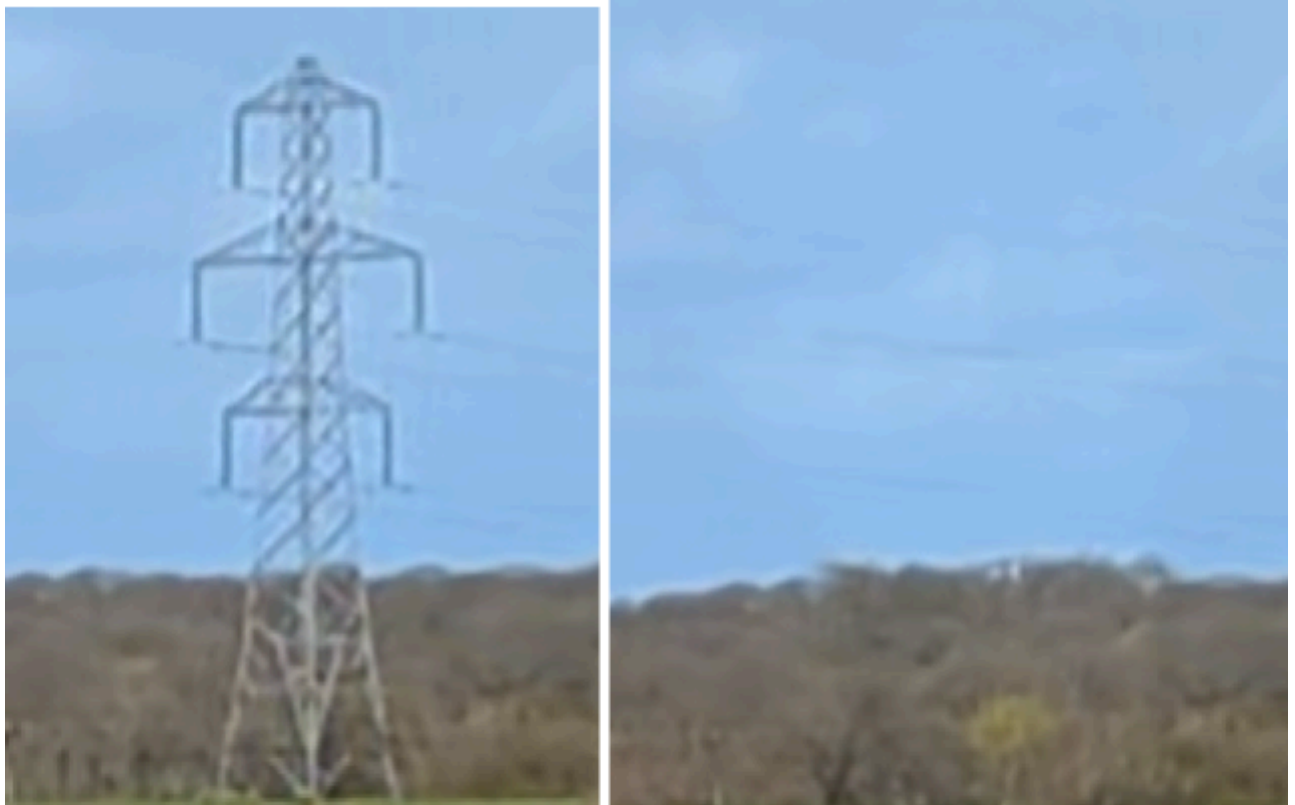


Figure SEQ Figure * ARABIC 47 Footage before Clone Stamp Figure SEQ Figure * ARABIC 47 Footage after Clone Stamp.

Conclusion

This section will assess the success of the project and whether it has achieved its aims and objectives. Suggestions for further development and improvement of the project will be suggested.

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Appendices

Log

- 06/10/2022 – Discussed upcoming deadlines including project proposal and interim log
- 13/10/2022 – Discussed initial ideas and decided on one to develop
- 20/10/2022 – Submitted project proposal and began to research visual effects methods
- 27/10/2022 – Drew storyboard and began writing interim report
- 03/11/2022 – Researched similar shots and continued with interim report
- 10/11/2022 – Submitted interim report

Project Proposal

Visual Effects Shot

By Ifan Edwards

230968

Supervisor - Richard Ellis

Visual effects allow filmmakers to extend the scope of their imagination beyond what is possible with practical special effects and on-set methods of capturing a scene. They have allowed writers to remove the limits they used to have to set on themselves while conjuring fiction from their minds and permitted artists to conceive fantastical worlds with no bounds. On the other end of the spectrum, visual effects have allowed us to see what long extinct creatures, or ruined castles, or ancient leaders may have looked like in their time.

Aims

In my project I aim to make a short video sequence containing a variety of production techniques including shot composure, lighting, sound, set dressing, green screens and costumes and post-production techniques including colour correction, audio balancing, keying, rotoscoping, 3D modelling, rigging, texturing, virtual lighting, and compositing. My scene will be of an earthquake causing a crack to form in the ground and revealing new areas below. I plan to have the crack emit light on to the real footage and will have to plan accordingly so that it looks realistic.

Primary Objectives

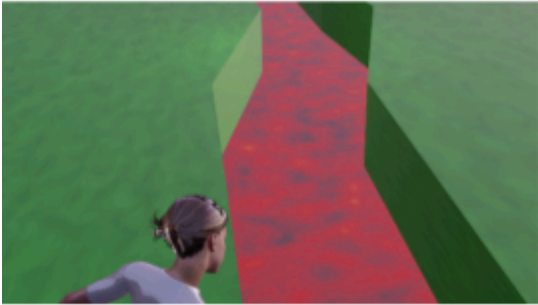
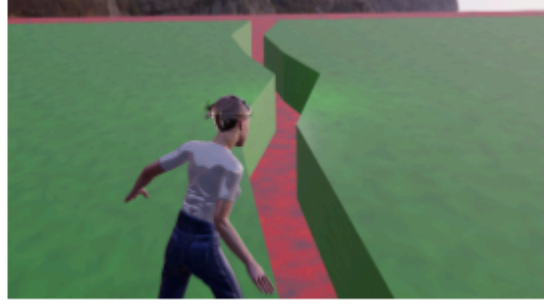
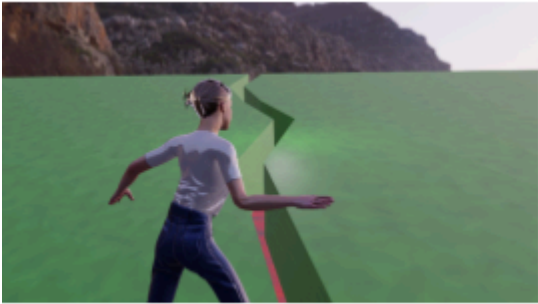
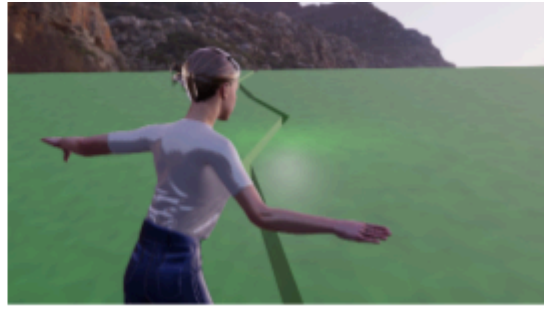
- Discover what details make visual effects convincing to the audience by analysing successful and unsuccessful visual effects shot created by studios in the filmmaking industry.
- Draw a storyboard for the scene so that the footage can be shot in as few takes as possible, saving time.
- Create a pre-visualisation of the scene as a plan for shooting the footage and rendering the 3D elements.
- Take an HDRI on set so that the lighting can be matched in the software, leading to more realistic highlights and shadows.
- Model the required assets (buildings, debris, lava etc.) so that they can be composited onto the footage.
- Create photoreal textures to apply to the 3D assets.
- Use camera tracking data from the real footage to use on the virtual camera in the computer-generated scene.

Extensions

- Use the physics built into 3D software to simulate destruction.
- Create an extended version of the shot.

Resources Required

- Use of the studio and green screens to film the footage.
- Professional camera to record highest quality footage possible.
- Blender, After Effects and a PC capable of rendering complex simulations.



User Testing Compliance Form

User Testing Compliance Form for UG and PGT Projects^{1*}

School of Engineering and Informatics

University of Sussex

This form should be used in conjunction with the document entitled “Research Ethics Guidance for UG and PGT Projects”.

Prior to conducting your project, you and your supervisor will have discussed the ethical implications of your research. If it was determined that your proposed project would comply with **all** of the points in this form, then both you and your supervisor should complete and sign the form on page 3, and submit the signed copy with your final project report/dissertation.

If this is not the case, you should refer back to the “Research Ethics Guidance for UG and PGT Projects” document for further guidance.

-
1. Participants were not exposed to any risks greater than those encountered in their normal working life.

Investigators have a responsibility to protect participants from physical, mental and emotional harm during the investigation. The risk of harm must be no greater than in ordinary life. Areas of potential risk that require ethical approval include, but are not limited to, investigations that require participant mobility (e.g. walking, running, use of public transport), unusual or repetitive activity or movement, physical hazards or discomfort, emotional distress, use of sensory deprivation (e.g. ear plugs or blindfolds), sensitive topics (e.g. sexual activity, drug use, political behaviour, ethnicity) or those which might induce discomfort, stress or anxiety (e.g. violent video games), bright or flashing lights, loud or disorienting noises, smell, taste, vibration, or force feedback.

2. The study materials were paper-based, or comprised software running on standard hardware.

Participants should not be exposed to any risks associated with the use of non-standard equipment: anything other than pen-and-paper, standard PCs, mobile phones, and tablet computers is considered non-standard.

3. All participants explicitly stated that they agreed to take part, and that their data could be used in the project.

Participants cannot take part in the study without their knowledge or consent (i.e. no covert observation). Covert observation, deception or withholding information are deemed to be high risk and require ethical approval through the relevant C-REC.

If the results of the evaluation are likely to be used beyond the term of the project (for example, the software is to be deployed, the data is to be published or there are future secondary uses of the data), then it will be necessary to obtain signed consent from each

^{1*}This checklist was originally developed by Professor Steven Brewster at the University of Glasgow, and modified by Dr Judith Good for use at the University of Sussex with his permission.

participant. Otherwise, verbal consent is sufficient, and should be explicitly requested in the introductory script (see Appendix 1).

4. No incentives were offered to the participants.
The payment of participants must not be used to induce them to risk harm beyond that which they risk without payment in their normal lifestyle. People volunteering to participate in research may be compensated financially e.g. for reasonable travel expenses. Payments made to individuals must not be so large as to induce individuals to risk harm beyond that which they would usually undertake.
5. No information about the evaluation or materials was intentionally withheld from the participants.
Withholding information from participants or misleading them is unacceptable without justifiable reasons for doing so. Any projects requiring deception (for example, only telling participants of the true purpose of the study afterwards so as not to influence their behaviour) are deemed high risk and require approval from the relevant C-REC.
6. No participant was under the age of 18.
Any studies involving children or young people are deemed to be high risk and require ethical approval through the relevant C-REC.
7. No participant had a disability or impairment that may have limited their understanding or communication or capacity to consent.
Projects involving participants with disabilities are deemed to be high risk and require ethical approval from the relevant C-REC.
8. Neither I nor my supervisor are in a position of authority or influence over any of the participants.
A position of authority or influence over any participant must not be allowed to pressurise participants to take part in, or remain in, any study.
9. All participants were informed that they could withdraw at any time.
All participants have the right to withdraw at any time during the investigation. They should be told this in the introductory script (see Appendix 1).
10. All participants have been informed of my contact details, and the contact details of my supervisor.
All participants must be able to contact the investigator and/or the supervisor after the investigation. They should be given contact details for both student and supervisor as part of the debriefing.

11. The evaluation was described in detail with all of the participants at the beginning of the session, and participants were fully debriefed at the end of the session. All participants were given the opportunity to ask questions at both the beginning and end of the session.

Participants must be provided with sufficient information prior to starting the session, and in the debriefing, to enable them to understand the nature of the investigation.

12. All the data collected from the participants is stored securely, and in an anonymous form.

All participant data (hard-copy and soft-copy) should be stored securely (i.e. locked filing cabinets for hard copy, password protected computer for electronic data), and in an anonymised form.

Project title: Earthquake Visual Effects Shot

Student's Name: Ifan Edwards

Student's Registration Number: 22001352

Student's Signature:



Date: 10/11/2022

Supervisor's Name: Richard Ellis

Supervisor's Signature:



Date: 09/11/22